

Does the Assessment Require the Skill?

Matching released mathematics keys without the tagged operation

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Abstract

Released mathematics items carry a published tag, a standard code or a cognitive domain, and a correct answer is read as evidence of that competency. If a pass does not require the tagged operation, that reading is of the wrong quantity. We ask whether a program lacking the operation can match the published key. A match is one-sided: it does not say students failed to learn, and finding none does not certify the tag.

On released items from nine authorities we score three Holm-corrected program families: item-writing rules, test-taking strategies, and language models shown the question with its given quantities withheld. The text layer of a released PDF often differs from the print, so claimed rows are measured on items re-read from the page and, for the language models, confirmed by a second independent transcription. The item-writing rules yield no family-wise discovery; one beats chance on two cells and awaits replication. Language models of two families recover New York Regents Algebra I and Algebra II keys, and the recovery survives a change of every option value with the form kept. On items that two independent codings of item demand mark as requiring computation on the givens, one of three scorers misses the bar; so on the matched items a pass does not require the given quantities, which is narrower than omitting the tagged operation. The international cognitive-domain cells, at about 50 to 80 public items each, yield no such program.

Keywords: assessment validity, test-wiseness, TIMSS

1 Introduction

An assessment item arrives with a claim about what answering it requires. Utah Core and the Common Core State Standards (CCSS) print a standard code. The Trends in International Mathematics and Science Study (TIMSS) prints knowing, applying, or reasoning [International Association for the Evaluation of Educational Achievement 2009]. A passing score is then used as evidence of that competency: in accountability, in curriculum evaluation, and in research that treats tagged items as operationalizations of skills. If a pass does not require the tagged operation, those uses are reading the wrong quantity. The testing standards treat that score use as a claim that must be validated for the intended interpretation [American Educational Research Association et al. 2014].

That inference is an argument, not a definition [Kane 1992, 2013; Messick 1995]. It can be wrong even when the publisher's tag is sincere. The item may be answerable from a

surface cue, from the choices without the stem, or from a formula that never binds symbols to the situation the standard names.

This paper searches for those rival explanations as programs. A program that lacks an operation the tag names, never sees the published key, and still matches the key under the published scoring rule is a *witness*. A witness refutes the claim that a pass on that item requires the tagged operation. A search that finds no witness does not certify the tag beyond the programs that were tried. We do not estimate whether students learned the skill, whether humans use the shortcuts, or which curriculum is better. We never say students do not learn X .

Results are pass rates of named programs on cells (an authority-by-tag group of items), not an item-by-operation table. The catalog rules lack every content operation. The language-model channel, which we call masked-stem, shows a model the question with every given quantity replaced by a placeholder, so what it lacks is information rather than an operation. TIMSS items keep their knowing, applying, and reasoning tags; we do not assign CCSS codes to TIMSS items.

Our contributions are the object, the bar, the census, and what the census found.

- **A one-sided test of a tag that needs no student responses.** We turn the claim that an item requires an operation into a falsifiable statement about programs: a program that lacks the tagged operation, never sees the published key, and matches the key under the published scoring rule is a witness that a pass does not require that operation. The test uses no student data and never yields a claim that students failed to learn. Validity theory states the argument; cognitive diagnosis and test-wiseness research estimate students or list cues. Neither scores this object (Section 2).
- **A pre-registered witness bar and family discipline.** A witness must fire on at least 10 items with a lower 95% confidence bound above format chance, adjusted for option elimination; searched and fitted programs must also beat the 95th percentile of random programs of matched complexity. Programs are hashed before scoring. Catalog rules, adaptive strategies, and partial-input language models are three families, each Holm-corrected on its own, so a row that beats chance in one cell is never billed as a family-wise discovery. Because the PDF text layer of a released form often differs from the printed page, every claimed row must also clear its bar on the item as printed: the

stem and options are re-read from the page image, and for the language-model rows a second independent transcription must agree with the first before the item enters the claimed population. Before any null is read, the same searcher must recover published shallow-cue rates on public benchmarks (Section 3).

- **A census of tagged public-release mathematics.** We assemble 2,405 keyed items from nine authorities, each tag quoted from its authority, including the last TIMSS cycle whose items, keys, and domain tags are public together. Channels are defined by the information and operations a program lacks: surface cues, adaptive test-taking strategies, substitution of the options into the printed equation read from the page image, and language models that see the question with every given quantity withheld.
- **What the census found.** The item-writing catalog yields no family-wise discovery; one rule, pick the second-smallest numeric option, beats chance on two cells (Ontario’s Education Quality and Accountability Office (EQAO) grade 6 and New York Regents geometry) on options re-read from the page, and awaits replication. The one strategy that survived correction failed a powered replication. Language models in two model families, Qwen2.5 and Phi-4, recover New York Regents Algebra I and Algebra II keys with the given quantities withheld, on items re-read from the page and verified by a second transcription, and the recovery survives a change of every option value with the option form kept, on all three scorers and both cells, so memorized items are not the whole effect. Two item-level codings of what each Algebra I item asks agree with each other moderately and with the publisher’s tag barely; on the items both coders mark as requiring operation on the givens, the two Qwen scorers clear the bar and Phi-4 does not. The claim is therefore about information, not operation: on the matched items a pass does not require the given quantities, which is narrower than a pass omitting the tagged operation. Substituting the options into the equation read from the page image answers a small subset of Algebra I. TIMSS knowing, applying, and reasoning stay null at about 50 to 80 public items per domain, a power-limited non-refutation, with the same searcher validated on benchmarks with documented cues (Section 4).

This census uses public-release items. TIMSS 2011 is the last cycle whose item texts, keys, scoring guides, domain tags, and percent correct are public together, with no permission process. From 2015, the International Association for the Evaluation of Educational Achievement (IEA) keeps most items secured; restricted-use grants typically cap example items and forbid the comparisons this census needs. Recency

of showable items comes from released state and provincial tests, not from a later TIMSS cycle.

2 Related Work

Standards labeling matches item text to a code. MathFish and related classifiers ask whether a labeler agrees with a publisher [Lucy et al. 2024]. Agreement between labelers is not a test of whether the tagged operation is required to pass. We keep the publisher’s tag and search for programs that pass without it.

Shortcut learning and partial-input baselines [Geirhos et al. 2020; Gururangan et al. 2018; Kaushik and Lipton 2018; McCoy et al. 2019; Poliak et al. 2018] show that models and rules can answer from the wrong slice of the input. Visual question answering (VQA) models can keep the same answer across images [Agrawal et al. 2016]. Choices-only language models already beat majority on several multiple-choice benchmarks; that success can be a surface artifact or abductive reconstruction of the question [Balepur et al. 2024]. This census is that construction on tagged public mathematics items, not the first choices-only result: the scorer sees the option set, and in the masked condition the question type, but not the given quantities, and we Holm-correct within a census that includes TIMSS cells. Later papers in the same line separate ranking artifacts from question inference and survey how multiple-choice evaluation is scored [Balepur and Rudinger 2024; Balepur et al. 2025]. A domain-naive classifier over stem and option features (length, stem to option and option to option similarity, keyword cues) trained to pick the keys of released science items reaches human-level guessing, and its above-chance accuracy is read as evidence that the item is a weak measure of the domain [Watson et al. 2018]; that is the closest program prior to our catalog and strategy families, which differ in using a priori hashed rules rather than a fitted classifier, tagged public mathematics, Holm correction within each family, and no student reference. Contamination of released items in training data is a rival explanation of those hits [Magar and Schwartz 2022; Sainz et al. 2023]; the Algebra I and Algebra II readers we claim have negative contamination probes and keep their rates above the channel bar when every option value is changed with the form kept (Section 4.3). Substituting the numerals in grade-school word problems lowers language-model accuracy [Mirzadeh et al. 2025]; that is the complementary experiment to withholding the given quantities, and we run that substitution on the option strings of the masked-stem Algebra I and Algebra II items as a control on the form reading. Item-writing and test-wiseness catalogs [Haladyna et al. 2002; Haladyna and Rodriguez 2013; Millman et al. 1965] list surface cues that can determine a choice without content, including using the numbers given without binding them. Those guidelines have been compiled into automatic item-writing-flaw detectors (longest option,

absolute terms, convergence, repeated words) [Moore et al. 2024]; that toolkit flags flaws rather than scoring pass rates, and the four extra catalog cues we scored after primary scoring overlap it. Some option sets already contain information about which member is the key even with the stem withheld [White and Zammarelli 1981]. Arithmetic closure is that unbound-execution move as a program. Those catalogs are our channels, not claims about students. Test makers and takers hide and seek keys in middle letter positions [Attali and Bar-Hillel 2003]; the lower-central program here picks a middle numeric rank, which is not that position cue. Plug-in substitution of numbered options is a documented SAT mathematics strategy [Katz et al. 2000]; the image-channel row is that strategy as a program on a displayed equation, not a student protocol.

Q-matrix and cognitive-diagnosis methods estimate which skills an item requires from student response data [de la Torre and Chiu 2016; Tatsuoaka 1983]. Expert attribute tags on TIMSS 2011 items have been treated as misspecifiable Q-matrix entries under the generalized deterministic inputs, noisy “and” gate (G-DINA) model [Terzi and Sen 2019]. Differential item functioning (DIF) uses examinee groups to detect items that do not measure the same construct across populations [Holland and Thayer 1986]. Language models have also been scored on the released TIMSS 2011 grade 8 mathematics items with item response theory against human populations [Fang et al. 2025]; that study asks how well models solve the items, whereas this census asks whether a partial-input program matches keys without the tag, so our TIMSS language-model null on the clean-option subset should be read next to that full-item result. This census is the student-free adversarial counterpart: no response vector is used. Comparisons to percent correct, when we make them, use only the statistics TIMSS already published with the public-release items.

Validity, cognitive diagnosis, test-wiseness, choices-only evaluation, and contamination detection each already study one piece of what a tagged correct answer can be taken to mean. Argument-based validity treats a tagged pass as a claim that must be supported for the intended use [American Educational Research Association et al. 2014; Kane 1992, 2013; Messick 1995]. Cognitive diagnosis and differential item functioning estimate attributes or bias from student responses. Test-wiseness catalogs list cues that can determine a choice without the tagged skill. Choices-only and contamination papers measure models on mostly untagged general-knowledge items. Domain-naïve guessers, item-writing-flaw detectors, psychometric benchmarking of models on TIMSS items, and numeral-substitution tests each occupy one vertex of that space [Fang et al. 2025; Mirzadeh et al. 2025; Moore et al. 2024; Watson et al. 2018]. Taken together they still do not score this paper’s object: a one-sided program witness on tagged public mathematics items, meaning a program that lacks the tagged operation, never sees the published key, and

still matches under the published scoring rule. A match is evidence that a pass does not require the tag. Failure to find a program does not certify the tag. Each family of programs is Holm-corrected on its own. That object is not a claim about whether students learned.

3 Method

3.1 Claims and corpora

Each claim is quoted from its authority. Utah Core grades 6–8 use Utah State Board of Education Core Guides. CCSS seed codes are checked against the same wording. TIMSS uses the 2011 framework definitions of knowing, applying, and reasoning [International Association for the Evaluation of Educational Achievement 2009].

Targets are released summative items and end-of-unit assessments with scoring rules. Practice problems are never targets. This census contains 2,405 keyed public items (2,245 selected-response, 160 numeric): New York Regents 1,448; State of Texas Assessments of Academic Readiness (STAAR) 445; Education Quality and Accountability Office (EQAO) 149; TIMSS 2011 129; earlier TIMSS public-release 106; Programme for International Student Assessment (PISA) 70; New York State Education Department (NYSED) 27; National Assessment Program Literacy and Numeracy (NAPLAN) 21; Massachusetts Comprehensive Assessment System (MCAS) 10. Operational Utah RISE items are secure and are not used. TIMSS 2015 and later are not used.

3.2 Channels

A channel fixes the information and the instruction set, and therefore the operations the program lacks. This census scores three families. The catalog family is the (rule, cell) list locked in the protocol draft; strategy programs were hashed before any pass rate was computed; the language-model family was scored the same day and Holm-corrected inside that family only. Hashes, timestamps, model revisions, seeds, and filter counts are in Appendix A.

Catalog rules. Item-writing cues (option length, position, absolute terms, and a signed variant of the catalog numeric-central-value cue) lack every content operation. Stem–option overlap and stem-number copying are shallow partial-input rules. Unbound execution applies a named formula (percent of, unit rate, $y = kx$, and the others in the 15-rule catalog) to stem numerals without binding symbols to the situation; it lacks bind and explain. Instruction-only (worked examples from the same unit) is specified but was not run: curriculum scoring keys were not in the public files we could ingest.

Adaptive strategy programs. Five programs, written from the same catalogs, were hashed and then scored; later parsing edits did not overwrite version-1 rates. A distractor-derivation hub picks the option from which the most other options can be derived under a fixed set of error transforms.

Constraint elimination removes options that a shallow stem keyword rules out (a negative count, a probability outside $[0, 1]$, a mean outside the listed data). Arithmetic closure applies $+$, $-$, $\times$, $\div$, squares, and square roots to the stem numbers to depth two and picks the option that first appears. Backsolving parses a one- or two-variable equation from the stem and picks the unique option that satisfies it. A fixed-order portfolio runs those four. None of the five binds a symbol to the situation the standard names. Chance for a rule that eliminates options is the mean of one divided by the number of remaining options, not 0.25. Hub and portfolio must also exceed the 95th percentile of random programs of matched complexity. Cells are every authority-by-claim pair with at least 10 selected-response items. A (program, cell) row enters the family when n scored is at least 10 and chance is positive: 67 tests. Holm is inside this family. A 172-test pool with the 105 catalog tests is a sensitivity, not a pool with the language-model tests. The catalog unbound formulas already scored in the 105 are a comparison, not this family’s closure program. Text-layer backsolving enters that family only when a cell has $n \geq 10$ unique fires; it does not. A later addendum, hashed before scoring and not a member of those 67 tests, uses a vision-language model (VLM) to transcribe the displayed equation from a page image, then substitutes each option. The model is not asked to solve. Invented formulas and extracts that do not match the page are abstentions. The addendum bar is n fired at least 10 and a lower 95% interval strictly above chance. Tie-breaks, the transform set, and the frozen hashes are in Appendix A.1.

Partial-input language models. A local instruction-tuned language model sees a slice of the item and picks the option letter with the highest next-token log-probability. There is no chain of thought and no free generation. Two model families, Qwen2.5 (7B-Instruct and 14B-Instruct) and Phi-4, are capable scorers; Mistral-7B-Instruct-v0.3 is a weak-solver check. Two prompts: options only (stem withheld entirely) and masked stem plus options. Masking replaces every numeral, number word, fraction, percentage, decimal, and value binding in the stem with the placeholder $[N]$; question wording, units, operators, and the option strings are kept. Items on which the masker withheld nothing are excluded from the primary population, because the tagged operation may still be performable; the primary population is the 1,215 items with at least one withheld quantity. The witness bar for this channel is: n at least 10, and the lower 95% bootstrap confidence interval strictly above both format chance and that cell’s modal-letter frequency, on the withheld-quantity population. A pass on those items does not require processing the givens. Contamination probes, not extra family tests, are mean token log-probability of the original unmasked stem plus options, administration-year buckets, option-letter permutation, and a second masking pass that also hides named figure types, variable bindings,

and leftover coefficients. A further pre-registered control, the option-numeral isomorph control, replaces every numeral in the option strings with a value-changing, form-preserving isomorph (same digit count, sign, and decimal places; one arm preserves the rank order of the options’ leading numbers and one scrambles it) and rescores the same masked stems, key letters, models, and letter rule; a rate that holds is a form effect, and a rate that falls to chance is not. Its thresholds, seed, and hashes are in Appendix A.2. Holm for this channel is every scored (model, cell) on the withheld-quantity masked-stem population with n at least 10: 68 tests (four models, 17 cells). Options-only is a ladder rung, not a member of that family. Holm is inside this family only. Catalog and strategy stay 105 and 67; the 172-test programmatic pool is a sensitivity and is not mixed with these 68. Model revisions, the letter-token rule, the clean-option filter, and the masking-rule hashes are in Appendix A.2.

3.3 Admission and the witness bar

Two kinds of catalog program are admitted.

1. *A priori rules*, taken from the catalogs with a citation each and no fitted parameters, scored on every item they apply to. There are 15 such programs (option length, position, absolute terms, a signed variant of the catalog numeric-central-value cue named below, stem-option overlap, stem-number copying, and unbound formulas).
2. *Fitted or searched programs*, trained on items disjoint from those they are scored on. Headline form is cross-corpus: fit on state items, score on TIMSS, or the reverse. Reported accuracy is held-out only.

Adaptive strategy programs and the language-model channel are separate families, pre-registered on the scoring day as above, not members of the 105. They do not replace the catalog family.

The catalogs list a numeric-central-value cue: among numeric options, the key is often a middle value [Haladyna et al. 2002; Millman et al. 1965]. The *a priori* program we score is a signed variant of that cue, which we call lower-central: parse a leading signed number from each option; if at least three parse, sort them and pick the option at index $\lfloor (n - 1)/2 \rfloor$. For four options that is the second-smallest number. The catalog median of four numbers is the average of the two central values, which is not an option, so a discrete program must pick one side. That index is fixed in the program; it is not a parameter fit on these pass rates. A protocol draft records option numeric rank (minimum, median, maximum) as searched features, not this exact *a priori* index. The same index is the named tie-break inside the strategy programs.

A catalog or strategy program is a witness on a claim only if n scored is at least 10, the lower 95% bootstrap confidence interval of the pass rate exceeds format chance (one divided by the number of options on that item, or the mean of one

divided by remaining options after elimination; 0.25 on four-option selected items with no elimination), and (for searched, fitted, hub, and portfolio programs) the pass rate is at or above the 95th percentile of random programs of matched complexity, which we call the machine bar. The partial-input language-model bar is stated in the paragraph above and is not loosened. A singleton match is logged, not claimed. Random-weight programs are the baseline, never witnesses.

The searched class is a library of surface-cue programs enumerated to bounded complexity, plus greedy and evolutionary search (Appendix A.3). Fourteen selected on the training corpus are scored on the other corpus only.

3.4 Three measurement layers and two codings of item demand

Every rate in this census was first computed on the PDF text layer of the released form. A 20-item hand audit against the printed pages found that layer to be dirty on about half of the sampled rows (Section 4.6). Two further layers follow.

Filter. For every item in the cells claimed on the text layer (955 items: the two algebra masked-stem populations, the two lower-central fired sets, and the unclaimed Phi-4 geometry cell), a vision-language model (Qwen2-VL-7B-Instruct, the model used for image backsolving) transcribes the stem and options of that item from the rendered page image. The model is never asked to solve or to pick a letter. Both the transcription and the text layer are normalized identically and compared token by token: the stem must reach token F1, precision, and recall of at least 0.8, every option must match at 0.9, and the option counts must agree; an item is then *print-faithful*. An unparseable transcription is *unverified* and is excluded from both subsets. The thresholds, prompt, and normalization were frozen and hashed before any population item was scored, after development on 40 items outside the population. The check was calibrated first on the 20 hand-audited items: 17 of 20 agree on the binary question (faithful or not), above the pre-registered floor of 16, so the thresholds were not adjusted. Every row claimed on the text layer was recomputed on the faithful subset and on its complement from the saved per-item flags and predictions, with the same bootstraps and the same bars. The faithful subset is not a random sample of the cell; it over-represents administrations before 2023, whose PDFs do not encode operators as digits, and items with word options, and a transcription error moves an item out of it and never in. Implementation details are in Appendix A.4.

Repair. The filter keeps a selected subset. The repaired layer replaces the text layer for every item in the two algebra masked-stem populations (606 items). The transcription above becomes the item’s stem and options; a second transcription of the same page, by the same model with a reworded prompt and a larger crop, is compared to the first under the thresholds above. An item on which the two

transcriptions agree is *repaired-verified*; every other item is *repaired-unverified*. The key letter is always the census key. The masked-stem, item-class, and option-isomorph rows are then scored again on the repaired stems and options with the same models, seed, and letter rule, on the verified population and on the full repaired population. The procedure, prompt, thresholds, and the amendment that re-runs Qwen 14B on the isomorph arms were hashed before the second transcription and before any scoring on this layer (Appendix A.6). Claims in the language-model channel are made on the repaired-verified population; the full repaired population is a sensitivity, and the text-layer and filtered rows are kept as an audit trail. For the two catalog rows, the stored transcription replaces the option strings where it exists, without a second transcription, and the rule is rerun.

Two codings of item demand. The masked-stem rows carry two codings of what the item asks. The first is the publisher’s tag. The NYSED rating guide of every Regents administration maps each question to a CCSS cluster, such as A-REI.B, so every item in the two algebra masked-stem populations has one. The verbs of each cluster heading are classified under a pre-registered rule as *execution* (solve, rewrite, factor, complete the square, create equations, compute, graph, perform arithmetic operations, and the like: operating on given quantities), *recognition or interpretation* (identify, interpret, recognize, compare, explain, understand), or *mixed* when the heading carries both. The rule and the classification of all 33 clusters were written and hashed before any prediction was joined to a tag (Appendix A.5). That classification is of the tag’s text, not of each item’s demand. The second coding is item-level: two language-model coders, GLM 5.2 and Kimi K3, each read every Algebra I item’s stem and options, with the key, every prediction, and the cluster split withheld, and assigned one of the same three codes under one fixed rubric (Appendix A.7). Agreement between the coders and between each coder and the tag is reported as Cohen’s κ . The masked-stem rows are recomputed on each tag class, on the items both coders give the same code, and on the intersection of those with the tag, at the unchanged channel bar. The population for a statement about the tagged operation is the items both coders mark as requiring operation on the givens; a statement about information withheld needs no such population.

3.5 Controls

Before a census null means anything, the same searcher has to find witnesses where they are known to exist. We run a synthetic Haladyna-flaw set (circular: the planted cues are the features) and, as the primary control, public multiple-choice corpora whose authors already published an options-only or question-ignored number [Lai et al. 2017; Mihaylov et al. 2018; Talmor et al. 2019; Zellers et al. 2018].

4 Results

Every rate in this census was first computed on the PDF text layer of the released form. That layer often differs from the printed page (Section 4.6), so the language-model rows and the two catalog rows that cleared on it were measured again on two further layers: a filter to the items whose text layer matches the page, and a repaired layer in which the stem and options are re-read from the page image. The repaired layer is the layer on which claims are made in this section; the filtered subset and the text layer are the audit trail (Appendix B).

4.1 Controls

On 1,200 synthetic four-option items, each planted atomic program scored 50/50 on holdout, confidence interval $[1, 1]$. Recall of planted flaw types is 6/6. False-positive rate on matched clean holdout is 0. That shows the searcher fires when a unique surface cue is planted. It does not show that real released items contain such cues.

On public cue sets, the same 1,847-program space, fit on a train split and scored held-out, is summarized in Table 1. The searcher recovers documented shallow cues on at least two corpora whose authors already published an options-only or question-ignored number. A TIMSS null is therefore not a failure of the tool to find anything.

4.2 Power

A claim cannot clear the bar below $n = 10$. For larger n , Table 2 records two floors. The independent-items floor is the smallest observed pass rate whose item-level bootstrap interval excludes chance, at the cell's tagged-item count. The clustered floor is the bypass rate with 80% power to clear that bar when administrations are resampled. Twenty-one cells in this census have $n \geq 10$. The smallest independent-items floor is 0.284, on New York Regents Algebra I ($n = 598$); that cell's clustered floor is 0.303. Public-release TIMSS items tagged knowing, applying, or reasoning pool the 2011 cycle with earlier released cycles at $n = 81/68/86$ tagged items and $n = 65/54/68$ selected-response items. Independent-item floors on the tagged counts are 0.358/0.353/0.372. Clustered 80% floors on the selected-response counts, from administration resampling in exports/addendum/power-by-cell.json, are 0.415/0.426/0.441. Those TIMSS cells are a null at n of about 50 to 80.

4.3 Partial-input language-model channel

A mechanical clean-option filter keeps 1,496 of 2,245 selected-response items, dropping exam footers, STAAR item codes, country-table fragments, options longer than 120 characters, and bleed from the next stem. TIMSS 2011 is mostly removed by country-table contamination (17 kept). Scorers take the letter with the highest next-token log-probability, with no chain of thought. The witness bar for this channel is $n \geq 10$ and a lower 95% interval strictly above both format chance

and the cell's modal-letter frequency, on items where at least one quantity was withheld. A pass on those items does not require processing the givens. It is not a claim that the model reasons, and not a claim that students guess.

The claimed rows in this channel are on the repaired layer of Section 4.6. For every New York Regents Algebra I and Algebra II item in the withheld-quantity population, the stem and options were re-read from the page image by a vision-language model, and a second, independent transcription of the same page was compared to the first under the print-fidelity thresholds. The 431 of 606 items on which the two transcriptions agree are the *repaired-verified* population; claims are made there. The full repaired population (*repaired-all*) is a sensitivity. The text-layer rows and the print-faithful filtered rows are kept in Appendix B as the audit trail. Other cells in this channel (geometry, EQAO, TIMSS, TEKS) were not re-read from the page, were measured on the text layer only, and none is claimed (Table 9).

Masked stem plus options. Every numeral, number word, fraction, percentage, decimal, and value binding in the stem is replaced by $[N]$. Question wording, units, and options remain. The primary population is the 1,215 of 1,487 clean scored items on which the masker withheld at least one quantity; Algebra I contributes 358 and Algebra II 248. On the repaired layer, 574 of those 606 items keep a usable transcription and at least one withheld quantity (Algebra I 343, Algebra II 231), and 261 and 167 of them are verified. Table 3 gives the rows. On the repaired-verified population Algebra I is $141/261 = 0.540$ $[0.479, 0.598]$ on Qwen 7B, $144/261 = 0.552$ $[0.494, 0.613]$ on Qwen 14B, and $129/261 = 0.494$ $[0.437, 0.559]$ on Phi-4, against chance 0.250 and modal B 0.287. Algebra II is $75/167 = 0.449$ $[0.371, 0.527]$ on Qwen 7B, $72/167 = 0.431$ $[0.353, 0.503]$ on Qwen 14B, and $69/167 = 0.413$ $[0.335, 0.491]$ on Phi-4, against chance 0.250 and modal C 0.275. All six rows clear both bars, and all six clear again on repaired-all (Algebra I $n = 343$, modal C 0.271; Algebra II $n = 231$, modal C 0.286). The claimed set is therefore Algebra I and Algebra II on Qwen 7B, Qwen 14B, and Phi-4.

Algebra II fails on the print-faithful filter and clears on the repaired layer. On the text layer it cleared on all three models (0.367, 0.359, 0.347 at $n = 248$). The print-faithful filter kept 115 of its 248 items, a subset selected toward administrations before 2023 and word options; there the point estimates (0.348, 0.383, 0.348) were close to the text-layer rates, but at half the n the lower bounds (0.261, 0.287, 0.270) were above chance and below that subset's modal-letter frequency 0.296. The filter had removed items, not the effect. The repaired layer scores 231 Algebra II items, 167 of them verified, and the interval clears the position baseline on every model. The repaired Algebra I rates (0.540, 0.552, 0.494 verified) are above the text-layer rates (0.388, 0.416, 0.422) on every model: the dirty layer deflated this row. On the repaired-verified population, one-sided binomial p against chance

Table 1. Method control: the same 1,847-program searcher recovers published options-only or question-ignored rates on OpenBookQA and SWAG. Chance is 0.25 except CommonsenseQA (0.20, five options). RACE and CommonsenseQA do not yield a witness here.

Corpus	Published cue number	Recovered	n	Witness
OpenBookQA	0.496, question ignored [Mihaylov et al. 2018]	0.450 (longest option)	329	yes
SWAG	0.436, ending-only [Zellers et al. 2018]	0.565 (searched)	62	yes
RACE [Lai et al. 2017]	(no official options-only number)	0.375	8	no ($n < 10$)
CommonsenseQA [Talmor et al. 2019]	chance 0.20	0.227	22	no

Table 2. Minimum detectable bypass rate at this cell’s n (every tagged item, not the subset a given rule scores). Independent is an item-level bootstrap on that n . Clustered is 80% power under administration resampling. TIMSS clustered entries are that resampling on selected-response items ($n = 65/54/68$ for knowing, applying, reasoning), not a rescaling of the independent floors. TIMSS rows pool 2011 and earlier public-release cycles.

Authority	Claim	n	Independent	Clustered
NY Regents	Algebra I	598	0.284	0.303
NY Regents	Algebra II	442	0.290	0.308
NY Regents	geometry	408	0.294	0.309
TIMSS	applying	68	0.353	0.426
TIMSS	knowing	81	0.358	0.415
TIMSS	reasoning	86	0.372	0.441
EQAO	grade 6	53	0.396	0.415

is below 10^{-16} for every Algebra I row and below 10^{-5} for every Algebra II row; the pre-registered bar is the bootstrap interval, and those p values are a description.

The language-model Holm family is every scored (model, cell) on the text-layer withheld-quantity masked-stem population with $n \geq 10$: 68 tests (four models, 17 cells). Options-only is a ladder rung, not a member of that family. Holm at $\alpha = 0.05$ in that family, against chance, rejects both Qwen Algebra I rows, both Qwen Algebra II rows, Phi-4 Algebra I, Phi-4 Algebra II, Phi-4 geometry, and Qwen 7B TEKS grade 8. The repaired layer adds no test to that family: it is a rescoring of the same two cells on a corrected text. Scorers are not averaged. Geometry on Phi-4 meets the text-layer cell bar ($67/166 = 0.404$) but not the split on items whose key is not the cell’s modal letter (0.229, $n = 118$, CI [0.161, 0.305]) and is not claimed; Qwen 7B clears TEKS grade 5 and grade 8 on that scorer only; Qwen 14B TIMSS knowing rejects Holm against chance and does not beat its population modal 0.409. Those text-layer rows and the options-only and information-ladder rungs are in Appendix B. One rater on the pre-registered 40-item Algebra I masked-stem sheet scored $15/40 = 0.375$ (bootstrap 95% CI [0.225, 0.525]); that is an uninformative null, not a human witness, and the claim is a model fact.

Which items carry the rows. Two codings of the items’ demand are joined to the repaired-verified rows (Table 4). The first is the pre-registered verb class of the publisher’s tag (Section 3.4): the CCSS cluster heading that NYSED assigns to the item names execution on given quantities, recognition or interpretation of a form, or both. On the repaired-verified Algebra I population the execution-tagged class ($n = 130$) clears on all three models (0.546, 0.515, 0.477 against modal B 0.323), as does the recognition-tagged class ($n = 105$: 0.514, 0.543, 0.495 against modal B 0.267). On the filtered subset the execution-tagged class had cleared on no model at $n = 75$; the repaired layer, with 130 execution-tagged verified items, reverses that. On Algebra II the execution-tagged class ($n = 83$) clears on all three and the recognition-tagged class ($n = 57$) clears on Qwen only (Phi-4 0.368 [0.246, 0.491]).

The second coding is item-level. Two language-model coders (GLM 5.2 and Kimi K3) each coded all 358 Algebra I items for what answering the item requires, under one fixed rubric with three codes, execution (the key requires operating on the specific given quantities), recognition or interpretation (the key can be picked from the form, meaning, or property named in the wording), or mixed, blind to the key, to every model prediction, and to the cluster split (Appendix A.7). The coders agree with each other moderately (Cohen’s $\kappa = 0.576$ on 356 items, observed agreement 0.789) and barely with the cluster split ($\kappa = 0.071$ and 0.230; observed agreement 0.461 and 0.564). Both coders see execution demand in more items than the tag does: 212 and 249 items coded execution against 164 execution-tagged, and of the 151 items tagged recognition or interpretation, 86 are coded execution by each coder and 79 by both. The cluster tag is therefore a noisy proxy for what the item asks, and we lean on the intersections. On the 142 verified items both coders code execution, Qwen 7B (0.401 [0.324, 0.486]) and Qwen 14B (0.437 [0.359, 0.521]) clear the bar against modal B 0.303; Phi-4 (0.352 [0.275, 0.430]) clears chance and not the modal-letter frequency. On the 79 items that both coders and the tag mark as execution, only Qwen 7B clears. On the 61 items both coders code recognition or interpretation, all three models clear at 0.738, 0.705, and 0.705 against modal A 0.295, and those rates survive the option-value change below (0.639, 0.689, 0.607).

Table 3. Masked-stem rows (given quantities withheld from the stem, options shown) on the two claimed cells, three scorers, three layers. Repaired verified: stem and options re-read from the page image and confirmed by an independent second transcription; this is the claimed population. Repaired all: every item with a usable page transcription, verified or not. Text layer: the PDF text layer as released. Chance is 0.250. Modal is the modal key-letter frequency of that population (verified: B 0.287 Algebra I, C 0.275 Algebra II; all: C 0.271, C 0.286; text layer: B 0.271, C 0.278). Bar: $n \geq 10$ and lower 95% CI above chance and above modal. Every row clears on every layer; the print-faithful filtered rows, on which Algebra II did not clear, are in Table 12.

Claim	Scorer	Repaired verified $n = 261 / 167$	Repaired all $n = 343 / 231$	Text layer $n = 358 / 248$
Algebra I	Qwen 7B	0.540 [0.479, 0.598]	0.504 [0.449, 0.560]	0.388 [0.341, 0.439]
Algebra I	Qwen 14B	0.552 [0.494, 0.613]	0.519 [0.464, 0.574]	0.416 [0.363, 0.469]
Algebra I	Phi-4	0.494 [0.437, 0.559]	0.493 [0.443, 0.545]	0.422 [0.374, 0.472]
Algebra II	Qwen 7B	0.449 [0.371, 0.527]	0.433 [0.368, 0.498]	0.367 [0.306, 0.423]
Algebra II	Qwen 14B	0.431 [0.353, 0.503]	0.416 [0.351, 0.481]	0.359 [0.298, 0.415]
Algebra II	Phi-4	0.413 [0.335, 0.491]	0.407 [0.342, 0.472]	0.347 [0.286, 0.407]

So the statement we make is the one about withheld information: on the matched items, a pass does not require the given quantities. The stronger statement, that on items requiring operation on the givens a pass does not require executing the tagged operation on them, would need the both-coders execution subset to clear on all three scorers on the verified population, and Phi-4 does not clear it; on repaired-all ($n = 188$) Phi-4 clears by 0.011 (lower bound 0.298 against modal C 0.287), which is not a margin to claim on. The execution-tagged cluster class does clear on all three, but both item-level coders read execution demand into 79 of the 151 recognition-tagged items and recognition into 18 to 41 of the 164 execution-tagged ones, so the tag class is not a population of execution items. The recognition-coded items carry the highest rates on every model.

Option-numeral isomorphs. The masking leaves the option strings untouched, and released Regents forms are widely mirrored, so a memorized item could carry the rate. The pre-registered control (Appendix A.2) rescores the same masked stems with every numeral in the option strings replaced by a value-changing, form-preserving isomorph: same digit count, sign, decimal places, and comma grouping, the same replacement wherever a literal repeats within the item, the rank order of the options’ leading numbers preserved, and the key letter unchanged. On the repaired layer the original and the perturbed letters were scored in the same run from the same model load, so no reproduction floor applies, and all three scorers are covered. Four items whose perturbation raised or timed out are excluded (n becomes 260 and 165 verified, 342 and 228 all); items whose options carry no numeral are unchanged and stay in the rate. Table 5 gives the rows. On the repaired-verified population every row clears the bar under the value change: Algebra I 0.500 [0.438, 0.562], 0.465 [0.404, 0.527], and 0.419 [0.354, 0.477] against modal B 0.288; Algebra II 0.412 [0.339, 0.485], 0.394 [0.321, 0.473], and 0.382 [0.315, 0.461] against modal C 0.273. The paired drop

on items with an option numeral is $-0.049 [-0.117, 0.022]$ for Qwen 7B, $-0.103 [-0.166, -0.040]$ for Qwen 14B, and $-0.090 [-0.157, -0.013]$ for Phi-4 on Algebra I ($n = 223$ paired), and includes zero for all three on Algebra II. So every scorer has a form component that survives a change of every option value, on both cells, and Qwen 14B and Phi-4 also have a numeral-specific component of about 0.1 on Algebra I. That component could be memorized option values or value cues that survive the masking, such as an option whose magnitude fits the question type; the control does not separate the two. The text-layer control, on which Qwen 14B was not scored because its re-scored letters fell below a reproduction floor, is kept as audit trail in Appendix B (Tables 14 and 15).

Contamination probes. The remaining probes are on the text layer. On the withheld-quantity items, 7B masked-stem accuracy by quartile of mean token log-probability of the original unmasked stem plus options is 0.349 in Q1 versus 0.439 in Q4 (Spearman 0.091; 14B Spearman 0.107, Q4 minus Q1 0.120). Recency on all 1,487 masked-stem 7B items is flat: 0.416 / 0.409 / 0.399 / 0.402 across before 2015, 2015 to 2019, 2020 to 2023, and 2024 and later ($n = 178 / 579 / 383 / 331$). Option permutation on the withheld-quantity items (7B) is 0.385 [0.357, 0.412] versus 0.376 unpermuted; Algebra I is 0.405 [0.355, 0.453]. Version-2 masking (named figures, bindings, leftover coefficients) on the version-1 withheld-quantity overlap is 0.370 versus 0.376, delta -0.007 (McNemar 50 version-2-only correct, 58 version-1-only). Extra masking does not create the channel. These probes and the isomorph control together do not support memorization of the items as the whole effect; they do not rule out a memorized share.

Scorer dependence. The claim is existential. A reader exists in the Qwen2.5 family (7B-Instruct and 14B-Instruct) and one outside it (Phi-4, a 14-billion-parameter instruct model) that recovers the key from question type plus option form on withheld-quantity Algebra I and Algebra II items re-read

Table 4. Masked-stem rows on the repaired-verified population by what the item asks, three scorers. Cluster tag: the pre-registered verb class of the CCSS cluster heading NYSED assigns to the item (execution on given quantities, or recognition or interpretation of a form). Both coders: the two blind item-level coders agree on that code. Bar: $n \geq 10$, lower 95% CI above chance 0.250 and above the modal key-letter frequency of that subset. The both-coders execution subset clears on Qwen only; the both-coders recognition subset clears on all three and carries the highest rates. Text-layer and filtered rows by cluster class are in Table 13.

Claim	Item class	n	Qwen 7B	Qwen 14B	Phi-4	Modal	Bar
Algebra I	cluster tag: execution	130	0.546 [0.454, 0.631]	0.515 [0.423, 0.600]	0.477 [0.385, 0.562]	B 0.323	all three
Algebra I	cluster tag: recognition or interpretation	105	0.514 [0.429, 0.610]	0.543 [0.448, 0.629]	0.495 [0.400, 0.590]	B 0.267	all three
Algebra I	both coders: execution	142	0.401 [0.324, 0.486]	0.437 [0.359, 0.521]	0.352 [0.275, 0.430]	B 0.303	Qwen only
Algebra I	both coders and tag: execution	79	0.430 [0.329, 0.544]	0.380 [0.278, 0.494]	0.354 [0.266, 0.456]	B 0.304	7B only
Algebra I	both coders: recognition or interpretation	61	0.738 [0.623, 0.836]	0.705 [0.590, 0.820]	0.705 [0.590, 0.820]	A 0.295	all three
Algebra I	both coders and tag: recognition or interpretation	35	0.657 [0.486, 0.800]	0.571 [0.429, 0.714]	0.657 [0.514, 0.800]	D 0.343	all three
Algebra II	cluster tag: execution	83	0.494 [0.386, 0.602]	0.422 [0.313, 0.530]	0.458 [0.349, 0.566]	D 0.301	all three
Algebra II	cluster tag: recognition or interpretation	57	0.474 [0.351, 0.614]	0.491 [0.368, 0.614]	0.368 [0.246, 0.491]	B 0.263	Qwen only

Table 5. Option-numeral isomorph control on the repaired layer. Every numeral in the option strings is replaced by a value-changing, form-preserving isomorph that keeps the rank order of the options’ leading numbers; the masked stem and the key letter are unchanged, so a rate that holds is read from the form of the options, not their values. Original and rank-preserving letters come from the same model load. Paired Δ is rank-preserving minus original on items with at least one option numeral ($n = 223$ Algebra I, 153 Algebra II, verified). Bar: $n \geq 10$, lower 95% CI above chance 0.250 and above the subset modal-letter frequency (verified B 0.288 and C 0.273; all C 0.272 and C 0.281). Every row clears on both populations.

Claim	Scorer	n	Original	Repaired verified			Bar	n	Repaired all			Bar
				Rank-preserving	Paired Δ	[95% CI]			Rank-preserving			
Algebra I	Qwen 7B	260	0.540	0.500 [0.438, 0.562]	-0.049	[-0.117, 0.022]	yes	342	0.471 [0.418, 0.523]		yes	
Algebra I	Qwen 14B	260	0.552	0.465 [0.404, 0.527]	-0.103	[-0.166, -0.040]	yes	342	0.430 [0.377, 0.482]		yes	
Algebra I	Phi-4	260	0.494	0.419 [0.354, 0.477]	-0.090	[-0.157, -0.013]	yes	342	0.409 [0.357, 0.459]		yes	
Algebra II	Qwen 7B	165	0.449	0.412 [0.339, 0.485]	-0.039	[-0.118, 0.046]	yes	228	0.395 [0.329, 0.456]		yes	
Algebra II	Qwen 14B	165	0.431	0.394 [0.321, 0.473]	-0.046	[-0.131, 0.046]	yes	228	0.377 [0.320, 0.439]		yes	
Algebra II	Phi-4	165	0.413	0.382 [0.315, 0.461]	-0.039	[-0.124, 0.039]	yes	228	0.351 [0.289, 0.408]		yes	

from the page. A weaker reader does not. Mistral-7B-Instruct-v0.3 (revision in Appendix A.2), scored on the text layer only, is 0.302 [0.257, 0.349] on Algebra I, which clears chance 0.250 but not modal B 0.271, and 0.262 [0.210, 0.319] on Algebra II, which clears neither; its with-stem letter-argmax ceiling matches its masked-stem rate (Algebra I 0.304, Algebra II 0.290; Table 11), so that miss is capability, not a missing cue. We do not average across scorers.

What the channel measures. Cue attribution on the text layer, Qwen 14B Algebra I: rate when the key is versus is not the lower-central option 0.341 ($n = 82$) versus 0.438 ($n = 276$); modal letter 0.454 ($n = 97$) versus 0.402 ($n = 261$); hub 0.370 ($n = 46$) versus 0.423 ($n = 312$); unique longest 0.333 ($n = 42$) versus 0.427 ($n = 316$); residual with none of the four cues 0.442 [0.368, 0.521], $n = 163$, versus chance 0.250. Option type: algebraic expressions 0.462 ($n = 65$), text 0.452 ($n = 188$), numeric 0.310 ($n = 71$), mixed 0.385 ($n = 26$), ordered pairs 0.250 ($n = 8$). Phi-4 on the same split: residual 0.380 [0.307, 0.460] ($n = 163$); algebraic 0.369, text 0.473, numeric 0.366 [0.254, 0.479] ($n = 71$, clears chance). The reader uses cues outside the catalog. On the items the model matches, a pass does not require the given quantities. Whether it requires the tagged operation is the question the

item codings above leave open: the both-coders execution subset clears on the two Qwen scorers and not on Phi-4.

An item with the quantities hidden. New York Regents Algebra I, January 2023, item 22, which NYSED maps to cluster A-APR. B, “understand the relationship between zeros and factors of polynomials” (a recognition-or-interpretation tag under the rule in Section 3.4); both item-level coders code it recognition or interpretation.

The item id is nyregents-algebra-i-2023-jan-q22.

Masked stem:

If the zeros of the function $g(x)$ are $\{[N], [N], [N]\}$, which function could represent $g(x)$?

- A. $g(x) = (x + 3)(x - 4)$
- B. $g(x) = (x - 3)(x + 4)$
- C. $g(x) = x(x + 3)(x - 4)$
- D. $g(x) = x(x - 3)(x + 4)$

Published key: C. On the repaired layer Qwen 7B and Qwen 14B pick C and Phi-4 picks D. The visible cue is the three placeholders in the zero set, which match the three-factor option that includes an x . That is form recognition of n factors from n placeholders, not execution on the given zeros. Here the tag itself names the zeros-to-factors relationship, so the match says the given zeros are not required; it does not say the tag is unread. The options are printed here as they

appear on the page. In the census text layer of this January 2023 form, =, +, and - are non-printing control bytes and the stem's minus sign is lost; the page transcription restores the operators. The two transcriptions of this item disagree (one drops the $g(x)$ = prefix from each option), so it is in the repaired-all population and not in the verified one.

4.4 Census

We ran 350,535 local trials. Fitted stem n -gram and overlap programs, state to TIMSS and the reverse, do not beat chance on holdout (state to TIMSS applying and reasoning: 36/122 = 0.295, interval [0.213, 0.377], chance 0.254). Unbound unit-rate, $y = kx$, and percent-of do not clear the bar on TIMSS, PISA, or NAPLAN numeric items.

No searched or a priori program cleared the bar on TIMSS knowing, applying, or reasoning. Failure to find a witness on public selected-response n of 65, 54, and 68 (clustered floors 0.415, 0.426, and 0.441; independent-item floors 0.353 to 0.372) does not certify those tags. It shows that this program space, on these public items, did not refute them.

Fifteen catalog rules were scored on every authority-by-claim cell. The a priori family is every (rule, cell) pair with n scored at least 10 and a positive format chance: 105 tests. One PISA numeric unit-rate row with chance 0 was excluded as degenerate. Five of those 105 tests clear the pre-registered per-cell bar on the text layer. None of the five survives Holm or Bonferroni at $\alpha = 0.05$ (Bonferroni cut 0.00048). There is no family-wise a priori discovery. Two of the five are the lower-central signed variant of the catalog numeric-central-value cue in Section 3, with one-sided binomial p values 0.0038 (geometry, 90/279) and 0.00070 (EQAO grade 6, 20/40) on the text layer. Table 6 gives those two rows on all three layers. On options re-read from the page image (Section 4.6) both clear the per-cell item bar: geometry 80/246 = 0.325, item interval [0.264, 0.382], and EQAO grade 6 18/38 = 0.474, item interval [0.316, 0.632]. On the filtered subset geometry had kept its rate and lost its interval at $n = 99$, and EQAO grade 6 had cleared at $n = 14$; the repaired options restore the n that the filter removed. Both rows are per-cell existence rows on the page-read options. Neither survives Holm over the 105 tests, and both remain candidates for replication rather than discoveries.

The fired n is the number of items on which the rule fires (at least three parseable option numbers): 279 of 408 New York Regents geometry items and 40 of 53 EQAO grade 6 items on the text layer, and 246 and 38 on the page-read options (two items in each cell have no stored transcription, and the geometry rule stops firing on 31 items whose transcribed options carry fewer than three numbers). The remaining items in those cells are not scored by this rule.

On the text layer, 19 of 27 geometry administrations have rate above chance (sign test $p = 0.014$); only one administration clears the item bar on its own. The rate is the same on filtered (0.323, $n = 99$) and not-faithful (0.326, $n = 178$)

items, so the dirt neither inflated nor deflated it; what the filter removed was n . On EQAO grade 6, all three years are above chance by rate; one year clears individually. The $k = 3$ clustered interval is descriptive. The text-layer geometry cue is central-ness: upper-central 0.330 ($n = 279$, item interval [0.272, 0.387]) and nearest-to-mean 0.312 also clear; smallest and largest both 0.219 do not. On EQAO grade 6 only the pre-registered lower-central index clears; upper-central is 0.250 (item interval [0.125, 0.400]). The EQAO witness is specific to the second-smallest option, not to a middle option in general.

The one-sided claim covers only the matched subset: the 80 of 246 geometry items and the 18 of 38 EQAO grade 6 items on which the rule returns the published key on the page-read options. On those items, matching the key does not require the tagged operation. The rule is a cell-level witness at the per-cell bar on the items it fires on. It is not a witness against each unmatched item in the cell.

The other three uncorrected clears are not claimed. New York Regents Algebra II, same lower-central program, is 0.298 at $n = 372$. The item interval is [0.253, 0.344] and the clustered interval is [0.255, 0.341]; the lower end sits on chance. Texas Essential Knowledge and Skills (TEKS) Algebra I stem-number repeat: 0.395, $n = 43$, interval [0.256, 0.535]. TEKS grade 5 unit-rate: 0.417, $n = 12$, interval [0.167, 0.750], chance 0.146. Searched programs that clear the machine bar on TEKS grade 6 and on Regents Algebra I / geometry have n between 11 and 20. We do not claim them: that n is the witness floor, 1,847 programs were searched, and all four searched programs fail a late-administration held-out test (late $n = 6$ to 10, intervals include chance). Fitted n -grams fail every temporal slice (state-pooled 0.257, $n = 1243$). One STAAR refit reaches 0.643 on $n = 14$ and is not claimed (train $n = 8$, one form, 1,562 programs searched).

Table 8 in Appendix B is every cell on which the lower-central rule scored $n \geq 10$ on the text layer. Over that 17-cell one-rule slice, Holm at $\alpha = 0.05$ keeps EQAO grade 6 and drops geometry. That slice is exploratory: it is the cells this one rule scored, defined after seeing which rule cleared, and the protocol names no primary family. TIMSS reasoning is 0.368, $n = 57$, interval [0.246, 0.491] there: the interval includes chance, so it is not a witness. Four further catalog cues were added after the primary scoring: longest option, convergence, stem-word overlap, and mean of other options. Hedge and grammatical never uniquely applied at $n \geq 10$. They give 22 further tests (127 total). The primary Holm family remains the 105 pre-registered tests. The combined family still has no Holm survivor. One further per-cell clear, not independent of the central-value cue, is mean of other options on geometry, 0.327 ($n = 156$, interval [0.256, 0.404]).

4.5 Adaptive strategy channels

The five strategy programs in Section 3 are a second family: 67 tests, hashed before scoring, Holm inside the family. Three

Table 6. Per-cell catalog existence rows for the lower-central rule (pick the second-smallest numeric option) on three layers: the PDF text layer, the print-faithful filter (items whose text layer matches the page), and options re-read from the page image (Section 4.6). n is items the rule fires on, not the cell size in Table 2. Item interval resamples items; clustered interval resamples administrations (year-session for Regents, year for EQAO). Bar: $n \geq 10$ and item lower bound above chance 0.250. Both rows clear on the page-read options and neither survives Holm over 105 a priori tests; they are replication candidates, not discoveries. The EQAO rule is specific to the second-smallest option.

Authority	Claim	Layer	n	Pass	Item 95% CI	Clustered 95% CI	Bar
NY Regents	geometry	text layer	279	0.323	[0.265, 0.380]	[0.265, 0.374]	yes
NY Regents	geometry	print-faithful filter	99	0.323	[0.232, 0.414]	[0.227, 0.414]	no
NY Regents	geometry	page-read options	246	0.325	[0.264, 0.382]	[0.264, 0.387]	yes
EQAO	grade 6	text layer	40	0.500	[0.350, 0.650]	[0.417, 0.588]	yes
EQAO	grade 6	print-faithful filter	14	0.571	[0.286, 0.857]	[0.400, 0.714]	yes, thin
EQAO	grade 6	page-read options	38	0.474	[0.316, 0.632]	[0.273, 0.588]	yes

rows clear the per-cell bar. A 172-test pool with the 105 catalog tests is a sensitivity: the same single Holm reject, and none of the original 105 become Holm rejects. Catalog, strategy, and language-model tests are three Holm families, never one mixed pool.

Arithmetic closure (unbound execution). On STAAR grade 5 the depth-2 closure fires on 40 of 60 selected items and matches 21 of those 40 keys. The rate is 0.525 ($n = 40$, item 95% CI [0.375, 0.675], chance 0.250, one-sided binomial $p = 0.000175$). That p is the smallest in the 67-test family (Holm threshold $0.05/67 = 0.000746$) and remains the only Holm reject in that family and in the 172-test programmatic sensitivity (threshold $0.05/172 = 0.000291$). It is a discovery-set Holm reject on the frozen 2019/2021/2022 census. It did not replicate on held-out administrations, below. The catalog unit-rate row on the same cell (0.417, $n = 12$, chance 0.146) was already scored and unclaimed; closure is a different program with larger n , not a second independent mechanism. Grade 5 is the only closure cell that clears on the discovery set. Geometry is 0.196 ($n = 224$, interval [0.143, 0.250]), which does not clear.

The registered rule still holds under the pre-registered robustness cuts: depth 1 only is $17/30 = 0.567$ [0.400, 0.733]; depth 2 without the geometry extras π and $ab/2$ is $20/40 = 0.500$ [0.350, 0.650]. Lower-central alone on the same 40 items is $13/40 = 0.325$ [0.200, 0.475] and does not clear. Random tie-breaks among minimum-depth options give expected rate 0.367 with a 1,000-draw interval [0.250, 0.475]; observed 0.525 sits above that interval. Twenty-seven of the 40 picks were settled by the lower-central tie-break among minimum-depth options (15 of those 27 correct); 13 were settled by depth alone. Mean closure density is 3.3 of 4 options. Clustered by STAAR year the interval is [0.455, 0.615]: 2019 8/13, 2021 8/16, 2022 5/11. The sign test on three administrations above chance is $p = 0.125$. Those three forms are the whole cell.

Thirteen of the 40 fired stems contain artifact numerals (axis ticks, thousands separators, stem-and-leaf digits, footer grade numbers, leaked neighboring text). That is a construct problem for those 13. The remaining 27, a population defined after the fact, are $16/27 = 0.593$ [0.370, 0.778]. Unique depth-1 closure (exactly one option in the depth-1 set) is 10 of 40 fired, and that unique option is the key on 6 of those 10; $n = 10$ is the bar edge. The frozen census row remains the Holm reject in the 67-test discovery family. It is not a confirmed family-wise discovery.

Replication on held-out STAAR grade 5. The registered prediction was that filtered arithmetic closure on new grade 5 items alone would have a lower 95% CI strictly above 0.250. On the version-1 unrepaired parse, 18 of 44 keys match (0.409, CI [0.273, 0.545]): the prediction is met, and $n = 44$ is below the registered power floor of 46. That positive is not described as underpowered. After a mechanical last-option page-leak repair, hashed before rescoring (42 last-option repairs, 3 stacked-fraction rejoins; hand-check 14 agree / 4 disagree / 2 uncertain, 0 key mismatches), 17 of 44 match (0.386, CI [0.250, 0.545]). The prediction is not met. That null is underpowered ($n = 44 < 46$). English grade 5 forms for 2017 and 2018 were then parsed (the 2015 key, the 2020 test, and the 2023 test were not found). New forms alone are 4 of 20 (0.200, CI [0.050, 0.400]). Combined repaired plus new is 21 of 64 (0.328, CI [0.219, 0.438]). Here $n = 64 \geq 46$. The prediction is not met, and this null is not underpowered. Grade 4 lower-central is still 10 of 17 after repair; 5 of those 10 hits are parse artifacts, and the clean 5 of 9 does not clear. It is not a witness. The Holm-surviving unbound-execution row on the frozen STAAR 2019/2021/2022 grade 5 cell did not replicate on held-out 2013/2014/2016/2017/2018 administrations after the pre-registered option-leak repair. Discovery 0.525, combined replication 0.328 with an interval that covers chance.

Hub, elimination, portfolio. The distractor-derivation hub never clears once the random-program 95th percentile is

required. Unique-hub share among 992 family-scored items is 0.189; most hubs are ties broken by lower-central. Geometry is 0.255 ($n = 161$, interval $[0.193, 0.317]$) versus chance 0.250 and p95 0.311. EQAO grade 6 is 0.321 ($n = 28$) versus p95 0.438. TEKS grade 4 is the only chance-only clear (0.455, $n = 22$, interval $[0.273, 0.682]$ versus p95 0.476) and is not a witness.

Constraint elimination barely eliminates: mean remaining options stay near 4 (geometry 3.92, EQAO grade 6 3.77), and the share of options removed is 0.025 on geometry and 0.019 on EQAO grade 6. Two specified intersections kill the published key (probability with percent-point options; interior-angle (0, 180) on a rectangle angle-sum). Key-survival is 0.985 on geometry, 0.906 on EQAO grade 6, and 0.877 on TEKS grade 8. A hand check of 15 keyword-matched items keeps 13 of 15 keys. The two per-cell clears, geometry 0.307 ($n = 404$, interval $[0.262, 0.351]$ versus effective chance 0.254) and EQAO grade 6 0.412 ($n = 51$, interval $[0.275, 0.529]$ versus 0.268), are the lower-central rule again. The keyword-matched geometry subset still clears (0.352, $n = 162$); the EQAO keyword subset ($n = 19$) does not. Those rows add nothing beyond the catalog lower-central rows on the same cells, which are judged on the page-read options above. TIMSS reasoning is 0.379 ($n = 66$, interval $[0.258, 0.500]$ versus effective chance 0.263) and does not clear.

The portfolio never clears the random-program 95th percentile (Table 7). EQAO grade 6 clears chance and one-over- k (0.453, $n = 53$, interval $[0.321, 0.585]$ versus 0.253) and still fails p95 0.472. Almost all mass is lower-central among survivors.

Image-channel back-solving. Text-layer back-solving still does not produce a 67-family row (6 unique fires among 1,186 solving-tagged stems; no cell $n \geq 10$). Most equations in the source PDFs are images. The image addendum was scored after that family and is not folded into those 67 Holm tests. Unique-satisfier fired on 17 of 598 Algebra I solving-tagged selected-response items; two fires are recoded as abstentions (an invented formula with no displayed equation; an incomplete two-equation extract). On the remaining 15, 14 match the key (0.933, $n = 15$, item 95% CI $[0.800, 1.000]$, chance 0.25). Coverage is $15/598 = 0.025$. The miss is a correctly transcribed equation on an item that asks which point is not on the graph. No other named cell reaches $n \geq 10$ fired. The row is an existence measurement after the catalog, like the EQAO lower-central row, not a family-wise discovery and not a census of Algebra I. On those 14 matched items a pass does not require solving. This row reads the printed equation from the page image, so it does not depend on the text layer.

Strategy controls. On the same OpenBookQA and SWAG hold-out splits used in Table 1, elimination and portfolio sit

on chance (OpenBookQA elimination 0.260, $n = 400$, interval $[0.218, 0.302]$; portfolio 0.268 $[0.225, 0.310]$; SWAG elimination 0.248 $[0.205, 0.288]$; portfolio 0.240 $[0.198, 0.280]$). Numeric hub, closure, and backsolve do not fire on those language items.

4.6 The text layer against the printed page

A 20-item printed-page audit, 10 geometry lower-central hits and 10 Algebra I masked-stem hits, compared each census row to the source exam PDF by hand (no optical character recognition). Nine rows match stem, options, and key. Eight have footer, figure-label, or previous-item glue on an option or stem. Two are truncated. One Algebra I row has the printed stem and a different option set (June 2023 item 3: printed lines $y = 2x - 2$ and three neighbors; census inequalities). An earlier count of 8 matches and 3 truncations included one row (January 2023 item 8) whose stem the audit tool had printed as a 400-character preview; the census stem is complete and matches the print. No sampled PDF was missing. Glue on geometry option D is usually a footer after a correct leading number.

Filter. The mechanical check of Section 3.4 then classified all 955 items in the claimed cells: 434 print-faithful, 502 not faithful, 19 unverified. The text layer is wrong in two main ways: glue from footers, diagram labels, table cells, or the next item (241 items), and option sets or stems that do not match the printed item (230), most of them Regents forms from 2023 onward whose PDFs encode $=$, $+$, and $-$ as the digits 5, 1, and 2. The faithful fraction per claimed cell is 0.52 (Algebra I), 0.48 (Algebra II), 0.36 (geometry lower-central), and 0.37 (EQAO grade 6). Table 12 in Appendix B recomputes every row claimed on the text layer on the faithful subset and on its complement. Algebra I masked-stem cleared on the 184 faithful items for all three models; Algebra II did not clear the modal-letter bar on its 115; geometry lower-central kept its rate and lost its interval on 99; EQAO grade 6 cleared on 14. The faithful subset is not a random sample of any cell: in Algebra I, 40 percent of faithful items have word options against 10 percent of not-faithful items, and administrations before 2023 dominate. It answers whether the effect exists on items whose text is what students saw; it does not estimate the rate on the cell, and a transcription mistake moves an item out of it, never in.

Repair. Filtering keeps a selected half of each cell. The repaired layer replaces the text layer instead. For all 606 Algebra I and Algebra II withheld-quantity items, the stem and options become the vision-language transcription of the page (transcription 1, the output already produced for the check above), and a second transcription of the same page is produced by the same model with a different prompt and a 6 pt larger crop (transcription 2). An item is *repaired-verified* when the two transcriptions agree under the print-fidelity thresholds (stem token F1, precision, and recall at least 0.8;

Table 7. Fixed-order strategy portfolio on the five headline cells. Rate is against effective chance of the branch used. Witnesses in this channel must also exceed the 95th percentile of 200 random programs of matched complexity. No cell clears that bar.

Authority	Claim	n	Pass	Eff. chance	95% CI	p95	Bar
NY Regents	geometry	408	0.279	0.253	[0.238, 0.321]	0.297	no
EQAO	grade 6	53	0.453	0.267	[0.321, 0.585]	0.472	no
TIMSS	knowing	65	0.308	0.271	[0.200, 0.431]	0.292	no
TIMSS	applying	54	0.333	0.254	[0.204, 0.463]	0.370	no
TIMSS	reasoning	68	0.309	0.263	[0.206, 0.426]	0.338	no

every option at least 0.9; equal option counts); the key letter is always the census key, never a model guess. The thresholds, prompt, and procedure were hashed before either the verification transcription or any scoring (Appendix A.6). Of 606 items, 431 are verified and 175 are not (Algebra I 263 and 95; Algebra II 168 and 80). Of the 431, 246 had been print-faithful and 185 had been not faithful under the filter: the repair recovers 185 items the filter had dropped, and 53 items the filter had kept do not verify. Fifteen items have no usable transcription 1 and 17 more have no withheld quantity in the transcribed stem; the withheld-quantity repaired population is 574 items, 343 Algebra I and 231 Algebra II, with 261 and 167 verified. The verified population is not selected on whether the text layer matched the page; its remaining selection is what two readings of the page agree on, and its verified fraction by cell is 0.73 and 0.68 rather than the filter’s 0.52 and 0.48. Every masked-stem, item-class, and isomorph row in Section 4.3 was then scored on that layer with the same models, seed, and letter rule. The two catalog rows use transcription 1 alone where it is stored (277 of 279 geometry and 38 of 40 EQAO grade 6 fired items): those options were not verified by a second transcription, and the rule re-fires on 246 and 38 of them (Table 6).

4.7 An item the lower-central rule answers

EQAO 2023 grade 6, released questions, printed card page 8 (key table page 21). Stem: “Which number has 4 as its tenths digit?”

A. 92.174 B. 85.043 C. 79.451 D. 43.256

Published key: C. File tag B1-KU: B1 is EQAO’s content code on this item; KU is the knowledge-and-understanding category. Item id eqao-2023-g6-q1.

The sorted numbers are 43.256, 79.451, 85.043, 92.174. The rule picks index $\lfloor (4 - 1)/2 \rfloor = 1$, which is 79.451, option C. The program never reads the stem and does not inspect tenths digits. Matching the key on this item does not require identifying a tenths digit.

This item is print-faithful under the check in Section 4.6, and its page-read options are the four numbers above. The geometry row is the same program on the multiple-choice session of the New York Regents geometry examination; on the 246 page-read fired items the rate is 0.325 with item

interval [0.264, 0.382], and on the 99 filtered items the same rate had an interval that covered chance.

EQAO grade 6 and NY Regents geometry are exam or grade labels, not TIMSS reasoning. We do not rank Ontario, New York, or TIMSS.

4.8 What would refute these claims

The EQAO grade 6 and geometry lower-central rows would be false if the published keys on the scored items were wrong, if the implemented rule picked any index other than $\lfloor (n - 1)/2 \rfloor$ on the sorted parsed option values, or if a recomputed lower 95% bootstrap confidence interval on the same correct/incorrect flags on the 38 and 246 page-read items did not exceed chance. Both rows are per-cell existence rows that no Holm family keeps; a family-wise miss is reported above and is not a refutation of the per-cell bar. The EQAO row is specific to the second-smallest option; an EQAO witness under upper-central or nearest-to-mean would change the specification claim on that row. The page-read options for these two cells come from one transcription without a second reading; a second transcription that disagreed on enough items to move the interval onto chance would remove the row.

The STAAR grade 5 closure discovery would be false if the published keys on those 40 fired items were wrong, if the implemented depth-2 closure with the registered tie-break did not produce 21 correct, or if the one-sided binomial p against chance 0.25 were larger than the Holm threshold on 67. That row is a discovery-set Holm reject that failed replication: the registered prediction on held-out filtered grade 5 items was a lower 95% CI strictly above 0.250, and the powered combined set is $21/64 = 0.328$, CI [0.219, 0.438]. Artifact numerals on 13 discovery stems and the three-administration sign test remain caveats on the discovery construct.

The Algebra I and Algebra II masked-stem claims would be false if a recomputed lower 95% interval on the repaired-verified withheld-quantity items did not exceed both chance and that population’s modal-letter frequency on that scorer, if the masking left a computable given quantity in the transcribed stem, or if the two page transcriptions that define the verified population were shown to share an error that put the key into the stem. A Holm miss against chance in the 68-test text-layer family would remove the family-wise

statement; it would not by itself remove the per-cell channel bar if the interval still cleared. Mistral failing the same bar does not refute the Qwen or Phi-4 rows. A with-stem ceiling at the masked-stem rate, as on Mistral, is a capability miss, not a missing-cue miss. The option-numeral control tests the mechanism behind the claim rather than the claim: had the rank-preserving rate on the repaired-verified items fallen to chance, the original rate would not have been a form effect. It did not on any scorer or cell (Table 5). The claim does not extend to a statement about the tagged operation: the both-coders execution subset clears on Qwen 7B and Qwen 14B and not on Phi-4, and if a further coding or a third scorer cleared it on every scorer, the information-withholding statement in Section 4.3 would be replaced by the stronger one.

The Algebra I image-backsolve existence row would be false if honest n fired were below 10, if the lower 95% interval on those 15 flags were not strictly above chance 0.25, or if the 14 hits were invented formulas rather than displayed equations. A Holm statement about this row is not claimed.

A TIMSS knowing, applying, or reasoning witness would have been any catalog, strategy, or partial-input program that clears the bar stated for its channel, including the withheld-quantity language-model bar. No such row exists in this file. Public selected-response n is 65, 54, and 68. Bypass rates below the clustered floors in Table 2 would not have been seen at that n .

5 Limitations

We did not measure students. A witness is a fact about a key and a program. It is not a claim that students use the program.

The one-sided claim covers only the matched subset of each claimed cell: the 18 of 38 EQAO grade 6 items and the 80 of 246 New York Regents geometry items on which the lower-central rule returns the published key on the page-read options; the repaired-verified Algebra I and Algebra II items on which a masked-stem model returns the key; and the 14 of 15 Algebra I displayed-equation items on which unique substitution returns the key. A pass on an item the program does not match has not been shown to omit the tagged operation.

We did not exhaust the class of programs that lack the tagged operation. A later search can find a TIMSS witness this one missed. On the withheld-quantity population, no channel in this file clears TIMSS knowing, applying, or reasoning. Public selected-response n is about 50 to 80, and the clustered 80% floors are 0.415 to 0.441, so a bypass rate in the low thirties would not have been seen. That null is power-limited and is not a certification of the tags.

Instruction-only (worked examples from the same unit) is specified in the protocol and was not run: curriculum scoring keys were not in the public files we could ingest.

The measurement instrument for the page is a vision-language model. About half of the census text layer in the claimed cells is not what students saw: of 955 items, 434 are print-faithful, 502 are not, and 19 could not be verified. The language-model claims are therefore made on a repaired layer in which the stem and options of every Algebra I and Algebra II item are the model's transcription of the page, verified by a second transcription. Both transcriptions come from one model and one revision (Qwen2-VL-7B-Instruct), with a different prompt and crop; an error the model makes the same way under both prompts passes verification, and the check has no human reading of the 606 pages beyond the 20-item audit (17 of 20 agree with the mechanical filter). Of 606 items, 175 do not verify and are scored only in the repaired-all sensitivity, which clears the same bars; 15 have no usable transcription and are not scored on this layer. The two catalog rows use one transcription without a second reading. Four Algebra items whose option perturbation raised or timed out are outside the isomorph rows. The verified population is not a random sample of the cell either: it is the items two readings of the page agree on, 0.73 of Algebra I and 0.68 of Algebra II, and a hard-to-read item is more likely to be outside it.

The EQAO grade 6 and geometry catalog rows are per-cell existence rows on the page-read options (18/38, interval [0.316, 0.632]; 80/246, interval [0.264, 0.382]). Neither survives Holm over the 105 a priori tests. The EQAO row is specific to the pre-registered lower-central index, and under the earlier filter it had cleared on 14 items and failed a post-hoc stricter fidelity cut (5/10). Both are replication candidates.

STAAR grade 5 closure on the frozen 2019/2021/2022 cell is a discovery-set Holm reject ($21/40 = 0.525$) that did not replicate on held-out 2013/2014/2016/2017/2018 administrations after a pre-registered option-leak repair (combined $21/64 = 0.328$, CI [0.219, 0.438], $n \geq 46$). Thirteen of 40 discovery stems contain artifact numerals. The artifact-free discovery rate 16/27 is on a population defined after the fact.

Text-layer back-solving does not reach $n \geq 10$ fired on any cell. The image addendum measures it on 15 of 598 Algebra I solving-tagged items (coverage 0.025); that is not a census of Algebra I. The row is not a 67-family Holm test.

The language-model channel has the operations. What is withheld is information: the stem, or every given quantity. The inference is that the published key is determined by question type plus the option set, not that the model solved the tagged exercise and not that a student would match the key from that slice. The model is not a student. Items on which the masker withheld nothing are excluded from that claim; TIMSS knowing's first-run rate was partly those items. The Algebra I and Algebra II witnesses hold on Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, and Phi-4 on the repaired-verified population and on the full repaired population. Neither holds on Mistral-7B-Instruct-v0.3 on the

text layer, whose with-stem ceiling matches its masked-stem rate, so that miss is a weak solver; Mistral was not rescored on the repaired layer. Geometry on Phi-4 meets the text-layer cell bar but not the non-modal-key split and is not claimed; Qwen 7B clears TEKS grade 5 and grade 8 on that scorer only, and those rows are not claimed.

What the item asks is measured two ways and the two disagree. The publisher's cluster tag, classified by its heading verbs, puts 164 Algebra I items in an execution class, and that class clears on all three models on the repaired-verified population. Two item-level coders, both language models rather than humans, agree with each other at $\kappa = 0.576$ and with the tag at $\kappa = 0.071$ and 0.230 ; they code 212 and 249 items execution and read execution demand into 79 recognition-tagged items. On the 142 verified items both coders code execution, Qwen 7B and 14B clear and Phi-4 clears chance but not the modal-letter frequency. We therefore do not state that a pass on execution items omits the tagged operation; we state that on the matched items a pass does not require the given quantities. The coders' rubric is fixed and hashed, but the coders were not validated against human raters, and a human coding could move items between classes.

The option-numeral control on the repaired layer leaves a form component in all three scorers on both cells: every rank-preserving rate clears the bar. It also shows a numeral-specific component of about 0.1 on Algebra I for Qwen 14B and Phi-4 (paired drops -0.103 and -0.090 with intervals excluding zero) that the control cannot split between memorized option values and value cues that survive the masking. The earlier probes and this control together do not support memorization of the items as the whole effect; they do not rule out a memorized share. On the text layer the same control excluded Qwen 14B by a reproduction floor; that exclusion no longer bears on the claim, and the text-layer rows stay in the appendix as the record.

We did not ingest every public paper. English forms from 2019, 2021, and 2022 are the discovery STAAR census. Earlier English STAAR forms scored here are a replication set, not additional census years. Most NYSED 2018 graphic stems, TIMSS 2007, and later restricted TIMSS cycles are out of this file.

Constructed-response items are not headline cells. Scoring guides that reduce to a value are matching; explanations were not graded here.

References

Aishwarya Agrawal, Dhruv Batra, and Devi Parikh. 2016. Analyzing the Behavior of Visual Question Answering Models. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, Jian Su, Kevin Duh, and Xavier Carreras (Eds.). Association for Computational Linguistics, Austin, Texas, 1955–1960. doi:10.18653/v1/D16-1203

American Educational Research Association, American Psychological Association, and National Council on Measurement in Education. 2014. *Standards for Educational and Psychological Testing*. American Educational Research Association, Washington, DC.

Yigal Attali and Maya Bar-Hillel. 2003. Guess Where: The Position of Correct Answers in Multiple-Choice Test Items as a Psychometric Variable. *Journal of Educational Measurement* 40, 2 (2003), 109–128. doi:10.1111/j.1745-3984.2003.tb01099.x

Nishant Balepur, Abhilasha Ravichander, and Rachel Rudinger. 2024. Artifacts or Abduction: How Do LLMs Answer Multiple-Choice Questions Without the Question?. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 10308–10330. doi:10.18653/v1/2024.acl-long.555

Nishant Balepur and Rachel Rudinger. 2024. Is Your Large Language Model Knowledgeable or a Choices-Only Cheater?. In *Proceedings of the 1st Workshop on Towards Knowledgeable Language Models (KnowLLM 2024)*, Sha Li, Manling Li, Michael JQ Zhang, Eunsol Choi, Mor Geva, Peter Hase, and Heng Ji (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 15–26. doi:10.18653/v1/2024.knowllm-1.2

Nishant Balepur, Rachel Rudinger, and Jordan Boyd-Graber. 2025. Which of These Best Describes Multiple Choice Evaluation with LLMs? A) Forced B) Flawed C) Fixable D) All of the Above. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (Eds.). Association for Computational Linguistics, Vienna, Austria, 3394–3418. doi:10.18653/v1/2025.acl-long.169

Jimmy de la Torre and Chia-Yi Chiu. 2016. A General Method of Empirical Q-matrix Validation. *Psychometrika* 81, 2 (2016), 253–273. doi:10.1007/s11336-015-9467-8

Qixiang Fang, Daniel Oberski, and Dong Nguyen. 2025. PATCH! Psychometrics-Assisted BenCHmarking of Large Language Models against Human Populations: A Case Study of Proficiency in 8th Grade Mathematics. In *Proceedings of the Fourth Workshop on Generation, Evaluation and Metrics (GEM²)*, Association for Computational Linguistics, Vienna, Austria and virtual meeting, 808–823. https://aclanthology.org/2025.gem-1.68/

Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. Shortcut learning in deep neural networks. *Nature Machine Intelligence* 2, 11 (Nov. 2020), 665–673. doi:10.1038/s42256-020-00257-z

Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. 2018. Annotation Artifacts in Natural Language Inference Data. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, Marilyn Walker, Heng Ji, and Amanda Stent (Eds.). Association for Computational Linguistics, New Orleans, Louisiana, 107–112. doi:10.18653/v1/N18-2017

Thomas M. Haladyna, Steven M. Downing, and Michael C. Rodriguez. 2002. A Review of Multiple-Choice Item-Writing Guidelines for Classroom Assessment. *Applied Measurement in Education* 15, 3 (July 2002), 309–333. doi:10.1207/s15324818ame1503_5

Thomas M. Haladyna and Michael C. Rodriguez. 2013. *Developing and Validating Test Items*. Routledge, New York. doi:10.4324/9780203850381

Paul W. Holland and Dorothy T. Thayer. 1986. Differential Item Functioning and the Mantel-Haenszel Procedure. *ETS Research Report Series* 1986, 2 (1986). doi:10.1002/j.2330-8516.1986.tb00186.x

International Association for the Evaluation of Educational Achievement. 2009. *TIMSS 2011 Assessment Frameworks*. Technical Report. TIMSS & PIRLS International Study Center, Boston College, Chestnut Hill, MA. https://timssandpirls.bc.edu/timss2011/frameworks.html

Michael T. Kane. 1992. An argument-based approach to validity. *Psychological Bulletin* 112, 3 (Nov. 1992), 527–535. doi:10.1037/0033-2909.112.3.527

Michael T. Kane. 2013. Validating the Interpretations and Uses of Test Scores. *Journal of Educational Measurement* 50, 1 (2013), 1–73. doi:10.1111/jedm.12000

- Irvin R. Katz, Randy Elliot Bennett, and Aliza E. Berger. 2000. Effects of Response Format on Difficulty of SAT-Mathematics Items: It's Not the Strategy. *Journal of Educational Measurement* 37, 1 (2000), 39–57. doi:10.1111/j.1745-3984.2000.tb01075.x
- Divyansh Kaushik and Zachary C. Lipton. 2018. How Much Reading Does Reading Comprehension Require? A Critical Investigation of Popular Benchmarks. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun'ichi Tsujii (Eds.). Association for Computational Linguistics, Brussels, Belgium, 5010–5015. doi:10.18653/v1/D18-1546
- Guokun Lai, Qizhe Xie, Hanxiao Liu, Yiming Yang, and Eduard Hovy. 2017. RACE: Large-scale Reading Comprehension Dataset From Examinations. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, Martha Palmer, Rebecca Hwa, and Sebastian Riedel (Eds.). Association for Computational Linguistics, Copenhagen, Denmark, 785–794. doi:10.18653/v1/D17-1082
- Li Lucy, Tal August, Rose E. Wang, Luca Soldaini, Courtney Allison, and Kyle Lo. 2024. Mathfish: Evaluating Language Model Math Reasoning via Grounding in Educational Curricula. arXiv:2408.04226 [cs.CL] <https://arxiv.org/abs/2408.04226>
- Inbal Magar and Roy Schwartz. 2022. Data Contamination: From Memorization to Exploitation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (Eds.). Association for Computational Linguistics, Dublin, Ireland, 157–165. doi:10.18653/v1/2022.acl-short.18
- R. Thomas McCoy, Ellie Pavlick, and Tal Linzen. 2019. Right for the Wrong Reasons: Diagnosing Syntactic Heuristics in Natural Language Inference. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, Anna Korhonen, David Traum, and Lluís Màrquez (Eds.). Association for Computational Linguistics, Florence, Italy, 3428–3448. doi:10.18653/v1/P19-1334
- Samuel Messick. 1995. Validity of psychological assessment: Validation of inferences from persons' responses and performances as scientific inquiry into score meaning. *American Psychologist* 50, 9 (Sept. 1995), 741–749. doi:10.1037/0003-066x.50.9.741
- Todor Mihaylov, Peter Clark, Tushar Khot, and Ashish Sabharwal. 2018. Can a Suit of Armor Conduct Electricity? A New Dataset for Open Book Question Answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun'ichi Tsujii (Eds.). Association for Computational Linguistics, Brussels, Belgium, 2381–2391. doi:10.18653/v1/D18-1260
- Jason Millman, Carol H. Bishop, and Robert Ebel. 1965. An Analysis of Test-Wiseness. *Educational and Psychological Measurement* 25, 3 (Oct. 1965), 707–726. doi:10.1177/001316446502500304
- Seyed Iman Mirzadeh, Keivan Alizadeh, Hooman Shahrokhi, Oncel Tuzel, Samy Bengio, and Mehrdad Farajtabar. 2025. GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models. In *The Thirteenth International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=AjXkRZlvjB>
- Steven Moore, Eamon Costello, Huy A. Nguyen, and John Stamper. 2024. An Automatic Question Usability Evaluation Toolkit. In *Artificial Intelligence in Education (AIED 2024), Lecture Notes in Computer Science*. Springer Nature Switzerland, 31–46. doi:10.1007/978-3-031-64299-9_3
- Adam Poliak, Jason Naradowsky, Aparajita Haldar, Rachel Rudinger, and Benjamin Van Durme. 2018. Hypothesis Only Baselines in Natural Language Inference. In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics*, Malvina Nissim, Jonathan Berant, and Alessandro Lenci (Eds.). Association for Computational Linguistics, New Orleans, Louisiana, 180–191. doi:10.18653/v1/S18-2023
- Oscar Sainz, Jon Campos, Iker García-Ferrero, Julen Etxaniz, Oier Lopez de Lacalle, and Eneko Agirre. 2023. NLP Evaluation in trouble: On the Need to Measure LLM Data Contamination for each Benchmark. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 10776–10787. doi:10.18653/v1/2023.findings-emnlp.722
- Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. 2019. CommonsenseQA: A Question Answering Challenge Targeting Commonsense Knowledge. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Jill Burstein, Christy Doran, and Thamar Solorio (Eds.). Association for Computational Linguistics, Minneapolis, Minnesota, 4149–4158. doi:10.18653/v1/N19-1421
- Kikumi K. Tatsuoka. 1983. Rule Space: An Approach for Dealing with Misconceptions Based on Item Response Theory. *Journal of Educational Measurement* 20, 4 (1983), 345–354. doi:10.1111/j.1745-3984.1983.tb00212.x
- Ragip Terzi and Sedat Sen. 2019. A Nondiagnostic Assessment for Diagnostic Purposes: Q-Matrix Validation and Item-Based Model Fit Evaluation for the TIMSS 2011 Assessment. *Sage Open* 9, 1 (2019), 2158244019832684. doi:10.1177/2158244019832684
- Patrick Watson, TengFei Ma, Ravi Tejjwani, Maria Chang, JaeWook Ahn, and Sharad Sundararajan. 2018. Human-level Multiple Choice Question Guessing Without Domain Knowledge: Machine-Learning of Framing Effects. In *Companion Proceedings of the The Web Conference 2018 (WWW '18 Companion)*. ACM, 299–303. doi:10.1145/3184558.3186340
- A. P. White and J. E. Zammarelli. 1981. Convergence Principles: Information in the Answer Sets of Some Multiple-Choice Intelligence Tests. *Applied Psychological Measurement* 5, 1 (1981), 21–27. doi:10.1177/014662168100500103
- Rowan Zellers, Yonatan Bisk, Roy Schwartz, and Yejin Choi. 2018. SWAG: A Large-Scale Adversarial Dataset for Grounded Commonsense Inference. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun'ichi Tsujii (Eds.). Association for Computational Linguistics, Brussels, Belgium, 93–104. doi:10.18653/v1/D18-1009

A Implementation details

This appendix records the settings that fix the reported numbers. Nothing here changes a claim in Section 3.

A.1 Frozen strategy programs

The hash of the frozen rule file, recorded 2026-09-16 20:36:16 UTC before any pass rate was computed, is a0eda4bdb02e54765bb7b49dca5e2e8ea9cd9ebb66de67b2aae074e7d555f0b4.

The distractor-derivation hub builds a directed graph on option values under a fixed error-transform set and picks the unique maximum out-degree, else lower-central among ties. Constraint elimination uses shallow stem keywords: nonnegative counts, probability in $[0, 1]$, percent in $[0, 100]$, positive magnitudes, angle ranges, and mean or median inside the listed data. Arithmetic closure at depth two is unbound execution of $+$, $-$, $\times$, $\div$, squares, and square roots on up to 12 stem numbers, with π and $ab/2$ on geometry-word stems; it picks the option that first appears, else lower-central among that depth. Backsolving parses a one- or two-variable equation from the stem and picks the unique option that satisfies it. The portfolio runs those four in a fixed order with lower-central among survivors and letter B last.

The machine bar for hub and portfolio is the 95th percentile of 200 random programs of matched complexity, seed 20260916.

The image-channel back-solving addendum was hashed before scoring; the sha256 of its preregistration is d6ee8c afd35d04bba7bfd4d4d364ee6f274d715e9b6b7020c454db dfbd994732. It uses Qwen2-VL-7B-Instruct (Hugging Face revision eed13092ef92e448dd6875b2a00151bd3f7db0ac) to transcribe the displayed equation from a page image; substitution is symbolic, outside the model.

A.2 Language-model scoring

The scored letter is the option letter with the highest next-token log-probability after the chat template, taking the maximum of the bare letter token and the space-prefixed letter token. The seed is 20260916.

Models and Hugging Face revisions:

- Qwen2.5-7B-Instruct: a09a35458c702b33eeacc393d103063234e8bc28
- Qwen2.5-14B-Instruct: cf98f3b3bbb457ad9e2bb7baf9a0125b6b88caa8
- Mistral-7B-Instruct-v0.3: c170c708c41dac9275d15a8fff4eca08d52bab71
- Phi-4 (microsoft/phi-4): 2db69c1c3e91a05d2c64a3185acfbaf36f744e25

The clean-option filter is mechanical. It keeps 1,496 of 2,245 selected-response items. Of those, 1,487 have a single option-letter key and are scored. The masker withheld nothing on 272 of those 1,487, leaving the primary population of 1,215 items with at least one withheld quantity. The first-run pooled rates on all 1,487 are a reference only.

The masking rules were hashed before scoring. Version 1 is af473bb6cd9d46dc74e034f4460d56a808273fe83ffa457cd7d122c40ae80e8. Version 2 is 8e6186728bf247d8349f1d94b68280b49eeebac7684c7ebeaf80f29345a43ae5.

Option-numeral isomorph control. The preregistration of this control was hashed before any scoring for it: 997242b9fb7437ffc2e86e2759e1373ba8ad98e231b0836ef8798a2266523e9b. Population: the 606 withheld-quantity Algebra I and Algebra II items under masking version 1 (358 and 248), with the subsets of Appendices A.4 and A.5. Seed 20260916; per item and arm the generator is seeded from the sha256 of seed, arm, and item id, so the substitution does not depend on processing order. Numerals in the option strings are integers (including coefficients and exponents), decimals, comma-grouped integers, and fractions. A replacement keeps the digit count, sign, number of decimal places, leading-dot form, and comma grouping, keeps denominators nonzero, and maps every repeat of the same literal within an item to the same new literal; zero becomes a positive one-digit integer. Variables, operators, units, words, option letters, letter order, and the key letter are unchanged. In the rank-preserving arm the leading numbers of the options are redrawn until

their pairwise order equals the original order, ties remaining ties; the scrambled arm has no order constraint. Items with no numeral in any option (61 Algebra I, 21 Algebra II) are unchanged under both arms, are included in pass rates, and are excluded from the paired difference. Example, June 2016 Algebra I item 7: options A to D 6.2, 7.3, 8.6, 8.8 become 7.4, 9.6, 9.7, 9.8 in the rank-preserving arm and 8.2, 4.9, 4.2, 2.7 in the scrambled arm. These invariants were asserted on every item before any perturbed arm was scored. Scoring reuses the masked-stem letter rule above, one model load per model. Original letters are re-scored, not copied, and must agree with the saved per-item predictions on at least 0.98 of the 606 ids before the perturbed arms are scored: Qwen2.5-7B-Instruct $595/606 = 0.982$ and Phi-4 $606/606 = 1.000$ cleared; Qwen2.5-14B-Instruct $592/606 = 0.977$ did not, so its perturbed arms were not scored. Pass-rate intervals are percentile bootstraps seeded by $21 + n$ and paired differences by $31 + n$, 1,000 replicates each. The pre-registered reading applies to Algebra I print-faithful items on the rank-preserving arm, each model separately: form-based if the perturbed rate clears the channel bar and the paired difference includes zero; not a form effect if the perturbed rate falls to chance; mixed otherwise. Under that reading, on the text layer, Phi-4 is form-based and Qwen 7B is mixed (Table 14). The same perturbation, rank-preserving arm only, was then applied to the repaired options of Appendix A.6 under the preregistration of that layer; there the original and the perturbed letters were scored in the same run, so no reproduction floor applies. On that layer Qwen 7B's paired difference includes zero on both cells and Qwen 14B's and Phi-4's exclude it on Algebra I (Table 5). Four repaired items whose perturbation raised or exceeded a 20-second limit are excluded from the perturbed arm and counted.

A.3 Searched program class

The feature library holds eleven surface cues, enumerated as conjunctions and signed weights up to complexity 3, plus greedy and evolutionary search: 1,847 programs logged.

A.4 Print-fidelity check

Population: the 955 items in the five cells named in Section 3.4 (Algebra I masked-stem 358, Algebra II masked-stem 248, geometry lower-central fired 279, EQAO grade 6 lower-central fired 40, geometry masked-stem 166; 136 geometry items are in both geometry cells). Transcriber: Qwen2-VL-7B-Instruct, revision eed13092ef92e448dd6875b2a00151bd3f7db0ac, bf16, greedy, 640 new tokens, seed 20260916, asked only to transcribe the stem and options of the numbered item from a 150 DPI render of the source PDF page (Regents: a crop from the question heading to the next heading; EQAO: the full page). The preregistration, with the prompt, the page locator, the normalization, and the thresholds, was hashed before any audited or population item was scored: 0211585

e9361d3771e04df5e118050ede761971da286f0b1333a7aac289737d6.

Normalization is identical on both sides: Unicode NFKC, LaTeX commands removed except trigonometric and logarithm names, subscripts and braces removed, unit words to symbols, a space at every letter-digit boundary, alphanumeric tokens, and the tokens for square root, π , percent, degree, and minus dropped. Control characters and operator glyphs therefore do not count; a digit standing in for an operator does. Stem score is multiset token precision, recall, and F1; option score is 1 for equal compact strings, else the maximum of token F1 and a character-level sequence ratio. Faithful: stem F1, precision, and recall each at least 0.8, every option at least 0.9, option counts equal. Unverified: unparseable output, no options, or empty stem. Calibration on the 20 hand-audited items: 17 of 20 agree on faithful versus not, against a pre-registered floor of 16 (the declared threshold grid gave 16 or 17 at every point), so the thresholds were kept. Census verdicts: 434 faithful, 502 not faithful, 19 unverified. Faithful fraction, faithful over faithful plus not faithful: Algebra I 0.523, Algebra II 0.481, geometry lower-central 0.357, EQAO grade 6 0.368, geometry masked-stem 0.657. Two post-hoc checks are disclosed and reported alongside the pre-registered subset: a strict variant that also drops any faithful item whose text layer carries a digit token absent from the transcription (85 of 434), and a locator repair that re-transcribed 12 Regents items and changed three verdicts; neither changes which rows clear, except that EQAO grade 6 does not clear on the strict subset (5/10). Rescoring reruns the catalog rule and reuses the saved per-item language-model predictions with the same bootstraps as the main text (item bootstrap seed 11, 1,000 replicates; cluster bootstrap seed 20260916, 2,000 replicates; language-model percentile bootstrap seeded by $21 + n$, 1,000 replicates). Wall time on one GPU: 26 minutes for the census.

A.5 Standards split

Tags: the “Map to the Learning Standards” table of the NYSED rating guide for each of the 58 Algebra I and Algebra II administrations in the masked-stem primary population, read from the PDF text layer (50 guides, 518 items), from a glyph table for three guides whose Calibri subset has a broken character map (30 items), or by template matching of rendered glyphs against six text-layer guides for five guides with no character map at all (57 items; 0 errors on 33,704 held-out characters). Every parsed table is validated (questions 1 to N contiguous, exactly 24 multiple-choice rows). All 358 Algebra I and 248 Algebra II items carry a cluster code; 33 distinct clusters are used. Cluster headings and standards were fetched from the CCSS site and stored with page hashes before classification. The verb-class rule, vocabulary, and classification of all 33 clusters were hashed before any prediction was joined to a tag: c87a5ae0d897b389dbf5cda52086c1754955f7cbb5f9e7a7382f5ac7cc475488.

Rule: each imperative verb heading a top-level clause of the cluster heading is execution (add, apply a rule, build, calculate, complete the square, compute, construct an equation or model, create, derive, evaluate an expression, factor, find, fit, graph, perform arithmetic operations, produce, rearrange, represent as a graph or table, rewrite, solve, summarize data, transform, translate between forms, write) or recognition or interpretation (analyze, choose, compare, construct an argument, decide, describe, distinguish, evaluate a process or claim, explain, identify, interpret, justify, know, make an inference, prove, reason, recognize, relate, understand). A purpose clause “use X to V ” takes the class of V ; “extend” and “use X ” with no purpose verb are unclassified, and a heading with no classifiable verb takes the class of its non-plus numbered standards. All execution: execution; all recognition: recognition or interpretation; both: mixed. Algebra I: execution 164 items (clusters A-APR.A, A-CED.A, A-REI.B, A-REI.C, A-REI.D, A-SSE.B, F-BF.A, F-BF.B), recognition or interpretation 151 (A-APR.B, A-REI.A, A-SSE.A, F-IF.A, F-IF.B, F-IF.C, F-LE.B, N-RN.B, S-ID.C), mixed 43 (F-LE.A, N-Q.A, S-ID.A, S-ID.B). Algebra II: 120, 93, and 35. A disclosed stricter execution class, which also requires that no numbered standard under the cluster be purely recognition, has 133 Algebra I items (66 print-faithful) and does not clear on any model on the text layer either. The text-layer and filtered rows use the saved per-item predictions and the same bootstrap as the main text; the full-cell and faithful rows reproduce the text-layer and print-faithful rows exactly. On the repaired layer the same class table is joined to the repaired scores: Algebra I verified 130 execution, 105 recognition or interpretation, 26 mixed; Algebra II verified 83, 57, and 27.

A.6 Repaired layer

The preregistration of the repaired layer and of the Qwen 14B amendment below was hashed before the verification transcription and before any scoring on that layer: 87ec5754c1b696b25366547793365610a13084f1230247b898f564ea9f92caee. Population: the 606 withheld-quantity Algebra I and Algebra II items (358 and 248). Transcription 1 is the stored output of the print-fidelity check (Qwen2-VL-7B-Instruct, revision eed13092ef92e448dd6875b2a00151bd3f7db0ac, prompt and crops of Appendix A.4); items whose stored output has no parseable stem or fewer than two options (15) are left without a repaired stem, and no text is invented. Transcription 2 is a second pass of the same model and revision at greedy decoding, 640 new tokens, seed 20260916, with two deliberate differences: the crop is expanded by 6 pt at the top and bottom (8 pt pad to 14 pt) and the prompt is reworded (it asks the model to copy the printed question, to skip headers, footers, other questions, and the computations box, to copy table cells row by row, to put no answer choice in the stem, and to return one JSON object). Parser, option-label map, and stem post-processing are the functions of the print-fidelity check. An item is repaired-verified when score_fidelity,

applied with transcription 1 as the item text and transcription 2 as the page reading, returns faithful (stem F1, precision, and recall each at least 0.8; every option at least 0.9; equal option counts). The key letter is the census key. Masking is version 1 (hash above) applied to the transcription-1 stem; options are not masked; the withheld-quantity population is items with a usable transcription 1, the census key among the transcribed options (no item fails this), and at least one masked token (17 items fail this).

Counts: 431 verified and 175 unverified of 606. Against the filter verdicts, verified items were 246 faithful and 185 not faithful; unverified items were 53 faithful, 107 not faithful, and 15 unverified. Algebra I: 263 verified (150 faithful, 113 not faithful) and 95 unverified; Algebra II: 168 verified (96, 72) and 80 unverified. Withheld-quantity population: 574 (343 Algebra I, 231 Algebra II), of which 261 and 167 verified. Scoring: the models, revisions, seed, prompt mode, and letter rule of Appendix A.2 (Qwen 7B batch size 4; Qwen 14B and Phi-4 batch size 1), on CUDA, one model load per model; the rank-preserving isomorph arm was scored in the same load. Intervals: random. Random. randrange percentile bootstrap, 1,000 replicates, seed $21+n$; paired differences seed $31+n$. Bar: $n \geq 10$, lower bound strictly above 0.25 and strictly above the subset modal-letter frequency. The preregistration’s honesty clause requires the README to say in its first section if a repaired-all Algebra I row does not clear; none fails. The catalog lower-central rows on the page-read options rerun the same rule on transcription-1 options where stored (item bootstrap seed 11, 1,000 replicates; cluster bootstrap seed 20260916, 2,000 replicates); no new transcription was made for those cells and no second transcription verifies them.

Qwen 14B amendment on the text layer. The original isomorph control excluded Qwen 14B at $592/606 = 0.977$ against a 0.98 reproduction floor. The preregistration above re-runs the 14B original masked-stem letters on the census text layer at batch size 1, keeps the floor at 0.98, and scores both perturbed arms whatever the agreement, with the label attached. Agreement was again $592/606 = 0.977$, label *below floor*; the 14 disagreeing item ids are listed in the export. Table 15 reports the arms so a reader can see them. It is not a pass of the control. The repaired-layer control in Table 5 covers Qwen 14B without a floor because the original and perturbed letters come from one run.

A.7 Item-level demand coding

Two language-model coders, GLM 5.2 and Kimi K3, each coded the 358 Algebra I withheld-quantity items under one fixed rubric with three codes: execution (to pick the key you must operate on the specific given quantities: solve, evaluate, compute, factor with these coefficients, rewrite this expression, create an equation from these numbers, graph from this equation, find this value), recognition or interpretation (you can pick the key by recognizing or interpreting the form,

meaning, or property named in the wording without computing on the given numbers), and mixed (needs both, or the key needs a small computation that any option-form reading would also give). Each coder received only the item id, stem, and options, with the key, every prediction, and every score field stripped by a helper script, and neither read the cluster split or the other coder’s output. Each coder’s README records the rubric verbatim with its sha256 (ae93b7c651614dd622bba6ff4a9eddb43c69b4b9ff67407d245365c18b11446 for the three rubric lines joined by newlines; 434ed271810ece24b0aa050e9424f0caaf7e3398d5654de7318a558505b6ef18 for the same lines as a bulleted text file). Item text came from the print-fidelity transcription where it existed (352 and 351 items) and from the text layer otherwise; the repaired layer did not yet exist when the coders ran. Coders could mark an item unreadable; GLM did so on two items whose options refer to a list absent from the transcribed text.

Codes: GLM 212 execution, 116 recognition or interpretation, 28 mixed, 2 unreadable; Kimi 249, 94, 15, 0; the cluster split 164, 151, 43. Cohen’s κ (unweighted, three classes, unreadable dropped): GLM versus Kimi 0.576 ($n = 356$, observed agreement 0.789); GLM versus cluster 0.071 (0.461); Kimi versus cluster 0.230 (0.564). GLM against Kimi: 195 items both execution, 82 both recognition or interpretation, 4 both mixed; 32 items GLM recognition and Kimi execution, 8 the reverse. Against the cluster tag, 86 of the 151 recognition-tagged items are coded execution by GLM and 86 by Kimi (79 by both), and 41 (GLM) and 18 (Kimi) of the 164 execution-tagged items are coded recognition or interpretation. Subsets used in Table 4: both coders execution 195 items (142 repaired-verified), both recognition 82 (61), both coders and the tag execution 97 (79), both coders and the tag recognition 51 (35). Rows are recomputed from the saved repaired-layer scores with the bootstrap above; the script and the per-subset id lists are in the export.

B Audit trail: text-layer and print-faithful rows

These tables are the rows on which earlier revisions made their claims, kept so a reader can follow each row from the text layer, through the print-faithful filter, to the repaired layer of Section 4.3. Nothing in this appendix is claimed on its own.

Options only. On the 1,487 scoreable clean items, both local Qwen models score 0.301 versus chance 0.250 and always-B 0.275 (7B interval [0.277, 0.326]; 14B [0.276, 0.323]). Letter agreement between models is 0.457. A with-stem sanity check on 7B is 0.460 on 200 items. A proprietary assistant scored through an audited no-tool transcript scores 0.359 ($n = 1493$, interval [0.334, 0.384]). No $n \geq 10$ cell has a 7B lower interval above that cell’s modal-letter frequency. Algebra I is 0.309 [0.265, 0.355] versus modal B 0.272 (111 of 408). On the 1,026 items whose key is not the cell modal letter,

Table 8. Lower-central on every cell with $n \geq 10$ (items the rule fires on), text layer. Geometry and EQAO grade 6 clear the per-cell bar here and again on the page-read options (Table 6). Algebra II clears the bar and is not claimed (lower end on chance). Holm on this 17-cell slice is exploratory. The other 88 of 105 a priori tests are omitted.

Authority	Claim	n	Pass	95% CI	Chance	Bar
NY Regents	geometry	279	0.323	[0.265, 0.380]	0.250	yes
EQAO	grade 6	40	0.500	[0.350, 0.650]	0.254	yes
NY Regents	Algebra II	372	0.298	[0.253, 0.344]	0.250	yes (not claimed)
NY Regents	Algebra I	458	0.260	[0.223, 0.301]	0.250	no
TIMSS	reasoning	57	0.368	[0.246, 0.491]	0.249	no
TIMSS	applying	50	0.240	[0.120, 0.360]	0.249	no
TIMSS	knowing	41	0.293	[0.146, 0.439]	0.248	no
TEKS	Algebra I	61	0.213	[0.115, 0.328]	0.250	no
TEKS	grade 6	52	0.327	[0.192, 0.462]	0.250	no
TEKS	grade 5	49	0.347	[0.224, 0.490]	0.250	no
TEKS	grade 8	45	0.267	[0.156, 0.400]	0.250	no
TEKS	grade 7	41	0.317	[0.171, 0.463]	0.250	no
TEKS	grade 3	36	0.278	[0.139, 0.444]	0.250	no
TEKS	grade 4	35	0.314	[0.171, 0.486]	0.250	no
EQAO	grade 9	38	0.211	[0.079, 0.342]	0.257	no
EQAO	grade 3	33	0.242	[0.091, 0.394]	0.253	no
NAPLAN	year 7	14	0.214	[0.000, 0.429]	0.250	no

7B is 0.318 [0.288, 0.347] versus chance 0.250. On 14B, Algebra I is 0.321 [0.279, 0.363] versus modal 0.272: the lower interval exceeds the modal frequency. That options-only Algebra I row is not a claimed witness; the claimed object is the masked-stem condition. Recency on the proprietary scorer minus 7B is largest before 2015 (+0.112, $n = 178$), then +0.024 in 2015–2019 ($n = 578$) and +0.069 in 2020 and later ($n = 712$).

Masked stem on the text layer. Of 1,487 scored items, 272 had nothing withheld; the primary population is the other 1,215. First-run pooled rates on all 1,487 (7B 0.405, 14B 0.425) mix those 272 in and are not the claim. Table 9 gives the named cells. Geometry does not beat its modal letter 0.289 on 7B (0.319, $n = 166$, interval [0.253, 0.392]) and does not clear chance on 14B (0.301 [0.235, 0.373]). EQAO grade 6 does not clear chance on 7B (0.400, $n = 40$, interval [0.225, 0.575] versus 0.255). TIMSS knowing on this population does not beat its modal letter 0.409 on 7B (0.500, $n = 22$, interval [0.318, 0.682]). The first-run TIMSS knowing rate 0.657 ($n = 35$) was carried by items where nothing was withheld: of 35 items, 18 are knowledge-answerable without any quantity, 12 are cue-readable from the leftover option set, and 5 are unclear. One-sided binomial p against chance on the text-layer withheld-quantity population: Algebra I 5.7×10^{-9} (Qwen 7B), 4.5×10^{-12} (Qwen 14B), and 9.4×10^{-13} (Phi-4); Algebra II 2.9×10^{-5} (7B), 9.0×10^{-5} (14B), and 4.2×10^{-4} (Phi-4); Phi-4 geometry 9.9×10^{-6} (67/166). Against the cell modal frequency: Algebra I 9.4×10^{-7} , 2.2×10^{-9} , and 5.6×10^{-10} ; Algebra II 0.00147, 0.00343, and 0.011; Phi-4 geometry 0.00104. Phi-4 text-layer rows: Algebra I 0.422 [0.374, 0.472]; Algebra II 0.347 [0.286, 0.407]; Algebra I residual with none of the four catalog cues 0.380 [0.307, 0.460] ($n = 163$); non-modal-key accuracy 0.387 ($n = 261$); numeric options 0.366 ($n = 71$); geometry 0.404 [0.331, 0.476] ($n = 166$) with non-modal-key split 0.229 ($n = 118$); pooled masked-stem 0.401

[0.374, 0.426] versus options-only 0.291 (McNemar 277/143). Qwen 14B on the 40-item human sheet is 0.425; on the 28 items whose census row matches the printed packet, Cohen’s κ with the rater is 0.212 (human 9/28, 14B 14/28).

Table 9. Partial-input language-model channel on named cells, Qwen and Mistral, withheld-quantity population, text layer. Chance is mean $1/k$. Modal is the cell’s modal key-letter frequency on that population. Bar: $n \geq 10$ and lower 95% CI above chance and above that modal frequency. Algebra I and Algebra II are claimed on the repaired layer (Table 3), not here. Geometry: Phi-4 meets the cell bar on the text layer but not the split on items whose key is not the cell’s most common letter and is not claimed. TEKS grade 5 and grade 8 clear on Qwen 7B only and are not claimed. Mistral-7B-Instruct-v0.3 does not clear.

Claim	n	Qwen 7B	Qwen 14B	Mistral 7B	Chance	Modal	Bar
Algebra I	358	0.388 [0.341, 0.439]	0.416 [0.363, 0.469]	0.302 [0.257, 0.349]	0.250	0.271	Qwen; not Mistral
Algebra II	248	0.367 [0.306, 0.423]	0.359 [0.298, 0.415]	0.262 [0.210, 0.319]	0.250	0.278	Qwen; not Mistral
geometry	166	0.319 [0.253, 0.392]	0.301 [0.235, 0.373]	0.295 [0.229, 0.373]	0.250	0.289	none
EQAO grade 6	40	0.400 [0.225, 0.575]	0.425 [0.300, 0.575]	0.150 [0.050, 0.275]	0.255	0.475	none
TIMSS knowing	22	0.500 [0.318, 0.682]	0.591 [0.409, 0.773]	0.364 [0.136, 0.545]	0.248	0.409	none
TIMSS applying	24	0.375 [0.208, 0.583]	0.375 [0.208, 0.583]	0.167 [0.042, 0.333]	0.248	0.333	none
TIMSS reasoning	42	0.286 [0.167, 0.429]	0.310 [0.167, 0.452]	0.238 [0.119, 0.381]	0.261	0.286	none
TEKS grade 5	38	0.474 [0.316, 0.632]	0.368 [0.211, 0.526]	0.184 [0.079, 0.316]	0.250	0.263	7B only; not claimed
TEKS grade 8	18	0.667 [0.444, 0.889]	0.611 [0.389, 0.833]	0.389 [0.167, 0.611]	0.250	0.389	7B only; not claimed

Table 10. Information ladder on text-layer withheld-quantity Algebra I ($n = 358$) and Algebra II ($n = 248$), Qwen 7B and 14B. Each rung adds one kind of information; the gap between masked-stem and with-stem is what the given quantities buy.

Claim	Model	Chance	Modal	Options	Masked	With stem
Algebra I	Qwen 7B	0.250	0.271 [0.226, 0.318]	0.313 [0.268, 0.360]	0.388 [0.341, 0.439]	0.508 [0.461, 0.559]
Algebra I	Qwen 14B	0.250	0.271 [0.226, 0.318]	0.332 [0.285, 0.380]	0.416 [0.363, 0.469]	0.575 [0.525, 0.626]
Algebra II	Qwen 7B	0.250	0.278 [0.222, 0.331]	0.274 [0.218, 0.335]	0.367 [0.306, 0.423]	0.403 [0.339, 0.464]
Algebra II	Qwen 14B	0.250	0.278 [0.222, 0.331]	0.286 [0.234, 0.343]	0.359 [0.298, 0.415]	0.440 [0.379, 0.500]

Table 11. With-stem letter-argmax ceiling versus masked-stem on text-layer withheld-quantity Algebra I ($n = 358$). Bar: lower 95% CI above chance 0.250 and modal B 0.271. Mistral’s ceiling is its masked-stem rate.

Scorer	With stem	Masked stem	Bar
Mistral 7B	0.304 [0.257, 0.352]	0.302 [0.257, 0.349]	chance only
Qwen 7B	0.508 [0.461, 0.559]	0.388 [0.341, 0.439]	both
Qwen 14B	0.575 [0.525, 0.626]	0.416 [0.363, 0.469]	both
Phi-4	0.497 [0.447, 0.553]	0.422 [0.374, 0.472]	both

Table 12. Every row claimed on the text layer, recomputed on print-faithful items and on the not-faithful complement (the filter step). Catalog rows: the lower-central rule; bar is item lower bound above chance. Language-model rows: masked-stem; bar is lower bound above chance 0.250 and above the modal key-letter frequency of that subset (Algebra I: 0.271 text layer, 0.255 faithful, 0.286 not faithful; Algebra II: 0.278, 0.296, 0.323). Algebra II cleared chance but not the modal-letter bar on its 115 faithful items and was dropped at this step; it re-enters on the repaired layer (Table 3). Geometry kept its rate and lost its interval at $n = 99$ and re-enters on the page-read options (Table 6).

Row	Scorer	n	Text layer			n	Print-faithful			n	Not faithful		
			Rate	[95% CI]	Bar		Rate	[95% CI]	Bar		Rate	[95% CI]	Bar
Geometry lower-central	rule	279	0.323	[0.265, 0.380]	yes	99	0.323	[0.232, 0.414]	no	178	0.326	[0.253, 0.399]	yes
EQAO grade 6 lower-central	rule	40	0.500	[0.350, 0.650]	yes	14	0.571	[0.286, 0.857]	yes	24	0.458	[0.250, 0.667]	no
Algebra I masked-stem	Qwen 7B	358	0.388	[0.341, 0.439]	yes	184	0.467	[0.391, 0.543]	yes	168	0.292	[0.220, 0.363]	no
Algebra I masked-stem	Qwen 14B	358	0.416	[0.363, 0.469]	yes	184	0.429	[0.359, 0.500]	yes	168	0.393	[0.321, 0.464]	yes
Algebra I masked-stem	Phi-4	358	0.422	[0.374, 0.472]	yes	184	0.467	[0.397, 0.538]	yes	168	0.363	[0.292, 0.440]	yes
Algebra II masked-stem	Qwen 7B	248	0.367	[0.306, 0.423]	yes	115	0.348	[0.261, 0.435]	no	124	0.395	[0.306, 0.476]	no
Algebra II masked-stem	Qwen 14B	248	0.359	[0.298, 0.415]	yes	115	0.383	[0.287, 0.461]	no	124	0.339	[0.250, 0.419]	no
Algebra II masked-stem	Phi-4	248	0.347	[0.286, 0.407]	yes	115	0.348	[0.270, 0.435]	no	124	0.347	[0.266, 0.427]	no

Table 13. Masked-stem rows by the verb class of the publisher’s tag on the text layer and on print-faithful items (the filter step). Bar: $n \geq 10$, lower 95% CI above chance 0.250 and above the modal key-letter frequency of that subset. On print-faithful items the execution-tagged Algebra I class ($n = 75$) cleared on no model; on the repaired-verified population ($n = 130$) it clears on all three (Table 4).

Claim	Tag class	n	Qwen 7B	Qwen 14B	Phi-4	Modal	Bar
Algebra I	execution, text layer	164	0.293 [0.220, 0.360]	0.378 [0.311, 0.457]	0.366 [0.293, 0.439]	B 0.299	14B only
Algebra I	execution, print-faithful	75	0.360 [0.253, 0.480]	0.333 [0.240, 0.440]	0.387 [0.293, 0.507]	B 0.293	none
Algebra I	recognition or interpretation, print-faithful	83	0.518 [0.410, 0.614]	0.446 [0.349, 0.554]	0.506 [0.398, 0.614]	D 0.265	all three
Algebra I	mixed, print-faithful	26	0.615 [0.423, 0.808]	0.654 [0.462, 0.846]	0.577 [0.385, 0.769]	C 0.346	all three
Algebra II	execution, text layer	120	0.392 [0.308, 0.483]	0.400 [0.317, 0.475]	0.367 [0.283, 0.450]	C 0.300	Qwen only
Algebra II	execution, print-faithful	56	0.393 [0.250, 0.518]	0.411 [0.286, 0.536]	0.339 [0.214, 0.464]	C 0.286	none
Algebra II	recognition or interpretation, print-faithful	38	0.316 [0.184, 0.474]	0.342 [0.211, 0.500]	0.316 [0.184, 0.474]	B 0.395	none
Algebra II	mixed, print-faithful	21	0.286 [0.095, 0.476]	0.381 [0.190, 0.571]	0.429 [0.238, 0.619]	A 0.333	none

Table 14. Option-numeral isomorph control on print-faithful Algebra I masked-stem items, text layer (the filter step). Original is the re-scored rate from the same model load; it differs from the saved rates by at most one item per row on Qwen 7B (11 of 606 letters changed on re-scoring) and by none on Phi-4. Rank-preserving keeps the order of the options’ leading numbers (primary arm); scrambled does not. Paired Δ is rank-preserving minus original on items with at least one option numeral ($n = 136, 56, 66$). Bar: lower 95% CI above chance 0.250 and above the subset modal-letter frequency (0.255, 0.265, 0.293), for the rank-preserving and the scrambled arm. Qwen 14B was not scored here: its re-scored original letters agreed with the saved letters on $592/606 = 0.977$, below the pre-registered floor of 0.98 (Table 15). The repaired-layer control that covers all three scorers is Table 5.

Scorer	Items	n	Original	Rank-preserving	Scrambled	Paired Δ [95% CI]	Bar
Qwen 7B	faithful	184	0.473 [0.397, 0.549]	0.391 [0.321, 0.462]	0.391 [0.321, 0.462]	-0.110 [-0.199, -0.015]	yes / yes
Qwen 7B	recognition or interpretation	83	0.530 [0.422, 0.627]	0.361 [0.253, 0.458]	0.410 [0.301, 0.518]	-0.250 [-0.375, -0.125]	no / yes
Qwen 7B	execution	75	0.360 [0.253, 0.480]	0.387 [0.280, 0.507]	0.320 [0.213, 0.427]	+0.030 [-0.106, 0.152]	no / no
Phi-4	faithful	184	0.467 [0.397, 0.538]	0.408 [0.342, 0.478]	0.446 [0.375, 0.516]	-0.081 [-0.169, 0.015]	yes / yes
Phi-4	recognition or interpretation	83	0.506 [0.398, 0.614]	0.398 [0.289, 0.506]	0.410 [0.301, 0.518]	-0.161 [-0.304, -0.036]	yes / yes
Phi-4	execution	75	0.387 [0.293, 0.507]	0.373 [0.267, 0.480]	0.440 [0.333, 0.560]	-0.015 [-0.152, 0.121]	no / yes

Table 15. Qwen 14B option-numeral arms on the text layer under the amendment of Appendix A.6. Re-scored original letters agreed with the saved letters on $592/606 = 0.977$, below the 0.98 floor, so this table is labelled below floor and is not a pass of the control. Paired Δ is rank-preserving minus original on items with an option numeral. Bar is for the rank-preserving arm against chance 0.250 and the subset modal-letter frequency.

Items	n	Original	Rank-preserving	Scrambled	Paired Δ [95% CI]	Bar
Algebra I, all	358	0.408 [0.355, 0.458]	0.360 [0.313, 0.413]	0.316 [0.271, 0.366]	-0.057 [-0.114, 0.000]	yes
Algebra I, faithful	184	0.418 [0.348, 0.489]	0.380 [0.315, 0.457]	0.370 [0.304, 0.446]	-0.051 [-0.132, 0.029]	yes
Algebra I, faithful, recognition-tagged	83	0.446 [0.337, 0.554]	0.386 [0.289, 0.494]	0.410 [0.301, 0.518]	-0.089 [-0.214, 0.054]	yes
Algebra I, faithful, execution-tagged	75	0.307 [0.213, 0.413]	0.333 [0.227, 0.440]	0.293 [0.187, 0.400]	+0.030 [-0.076, 0.136]	no
Algebra II, all	248	0.363 [0.302, 0.423]	0.323 [0.262, 0.379]	0.294 [0.238, 0.351]	-0.044 [-0.119, 0.026]	no
Algebra II, faithful	115	0.374 [0.287, 0.452]	0.339 [0.252, 0.435]	0.278 [0.200, 0.365]	-0.038 [-0.160, 0.085]	no